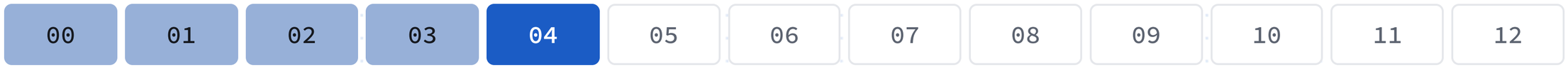
MODULE 04 · AMBITIOUS ENTREPRENEURSHIP

Validating Problems

What evidence would change your mind?

COURSE PATH



COURSE MATERIAL · SCAN NOW

All slides live on GitHub

The newest version, free, no account needed

github.com/BaNburger/
AmbitiousEntrepreneurship

1. **Open the link**
   github.com/BaNburger/AmbitiousEntrepreneurship
2. **Go to modules**
   then 04-validating-problems
3. **Click the lecture PDF**
   M04_Lecture_Validating-Problems.pdf, download top right

Public course repository of Ambitious Entrepreneurship. The QR code opens github.com/BaNburger/AmbitiousEntrepreneurship.

WHERE WE ARE · THE COURSE

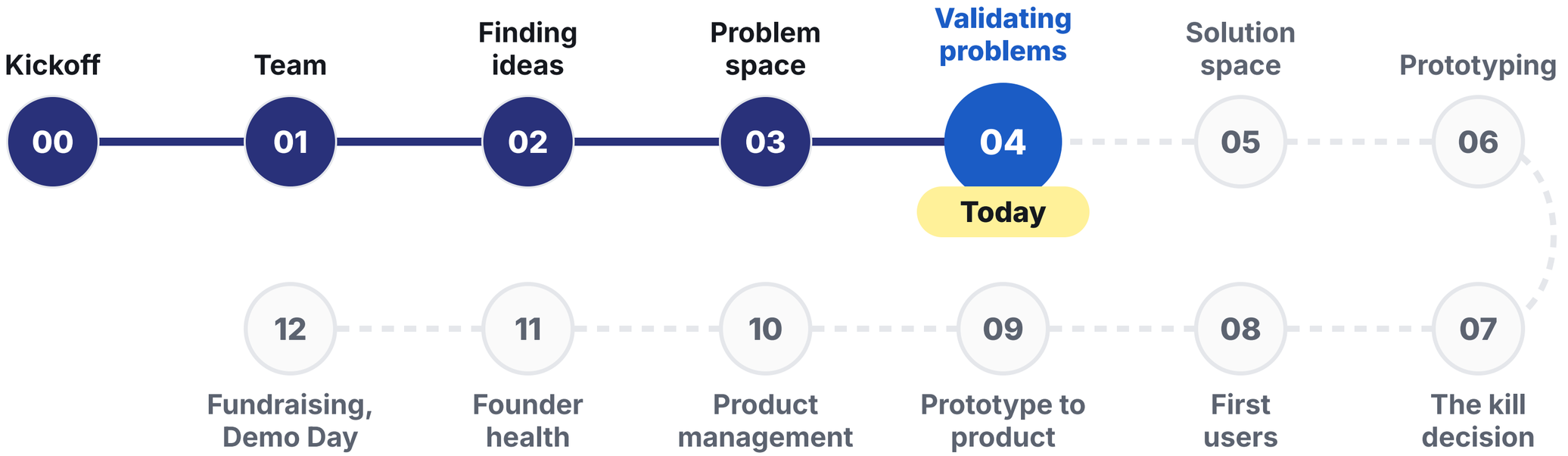
From what people say to what they do

Thirteen sessions from first idea to first revenue. What teams have by now:

One idea, picked on evidence
Finding ideas (M02)

A hair-on-fire segment
Problem space (M03)

Named people for a next step
Problem space (M03)



CHECK-IN · SHOW OF HANDS

How many people agreed to a next step?

Hands up for your band. One click on the band per hand



0

not yet



1

started

under the floor of 2



2–4

floor reached



5+

full bar

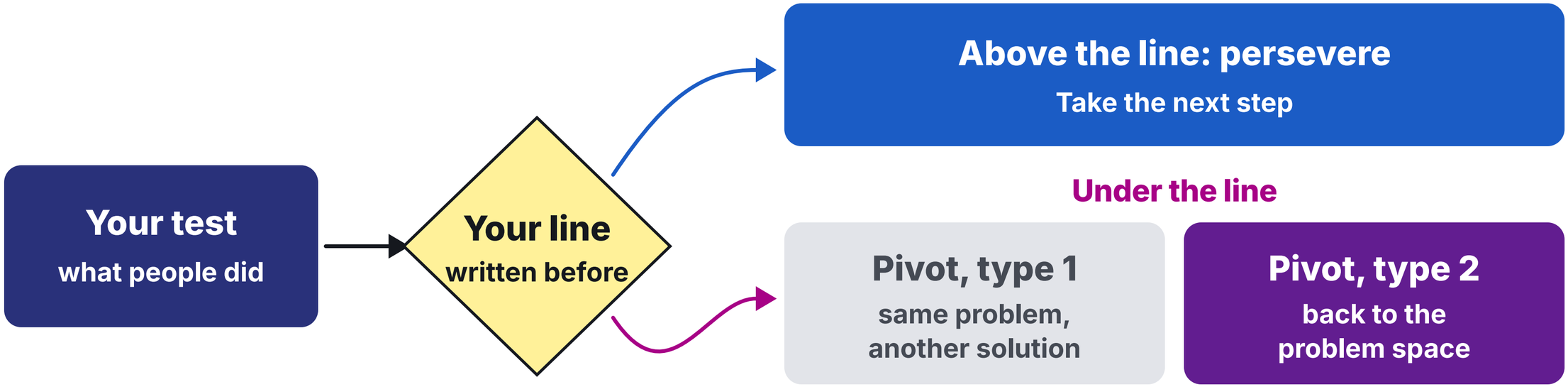
5 per student

2:00

PIVOT OR PERSEVERE

Under your line? That is a decision, not a failure

Before there is a solution, a pivot goes back to the problem



Both are progress. The only miss: no line, so no decision

FRAMEWORK Pivot or persevere after Ries, E. (2011), The Lean Startup. Pivot types as in the course Glossary.

ACCOUNTABILITY

Report what you did, not what you planned



User interactions?

What did you learn?



New revenue?

Why did they pay?



Shipped?

Why does it matter?

🦄 **Report target vs. actual in small groups**

CHAPTER 1 OF 9

Previously on Ambitious Entrepreneurship

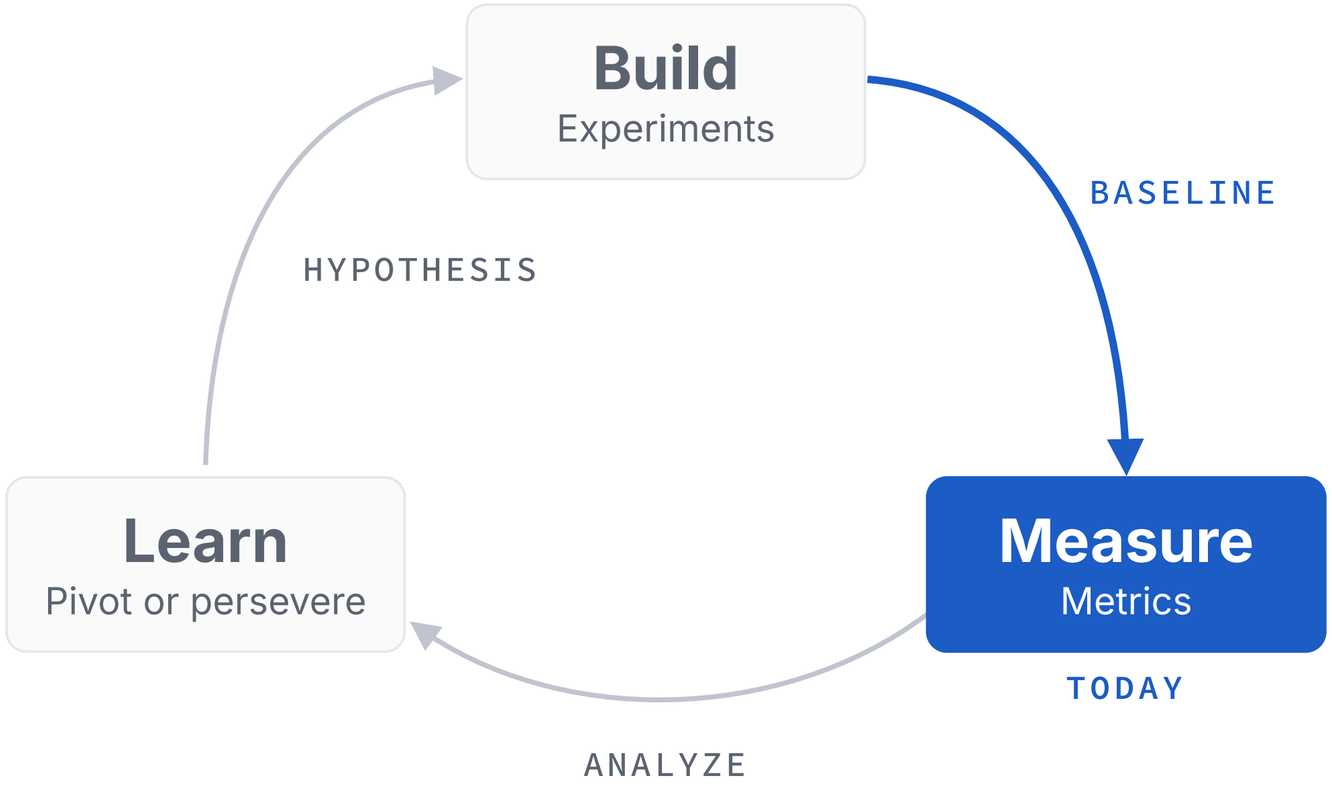
You heard who hurts, and from today you count who acts.

01 **Previously on Ambitious Entrepreneurship**
02 Your presentations
03 **Rank it: what your evidence proves**
04 **Draw it: the line before the data**
05 **Launch it: a page that tests your problem**
06 **Count it: visitors from your segment**
07 Wrap-up and check-out
08 Workshop: launch a page that tests your problem
09 Case discussion

PREVIOUSLY · KICKOFF (M00)

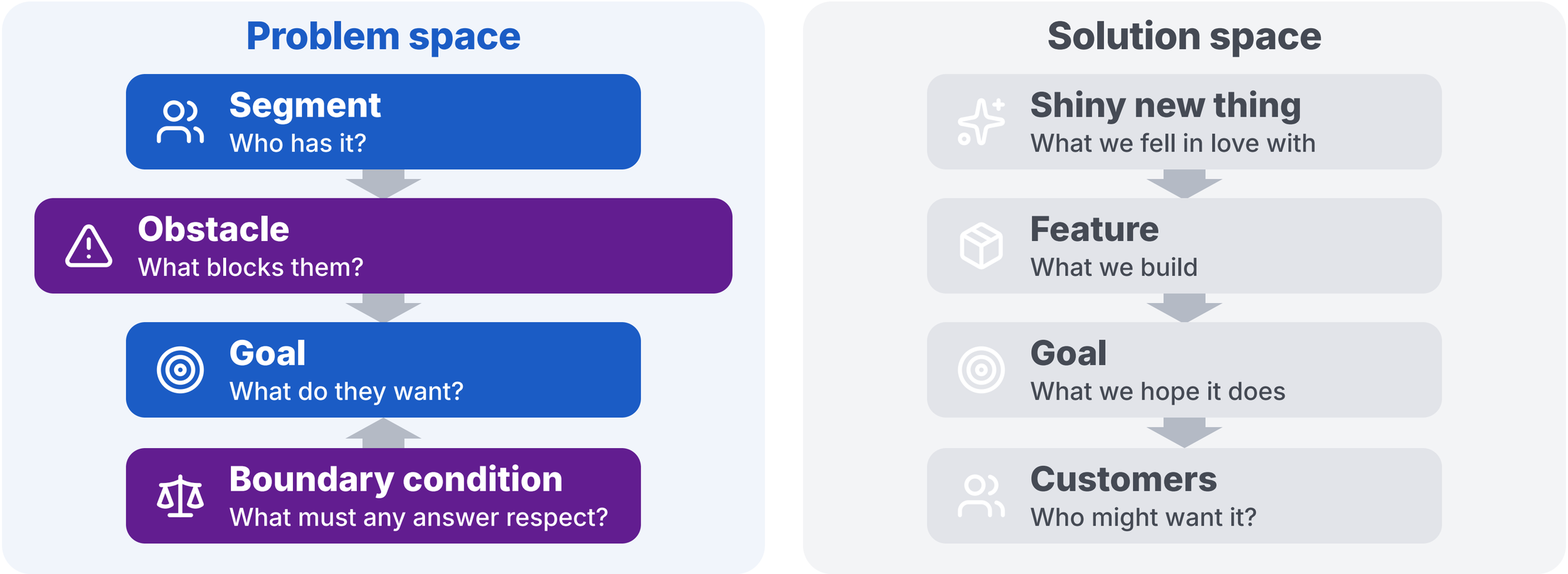
Your idea is a hypothesis

Today you measure the problem



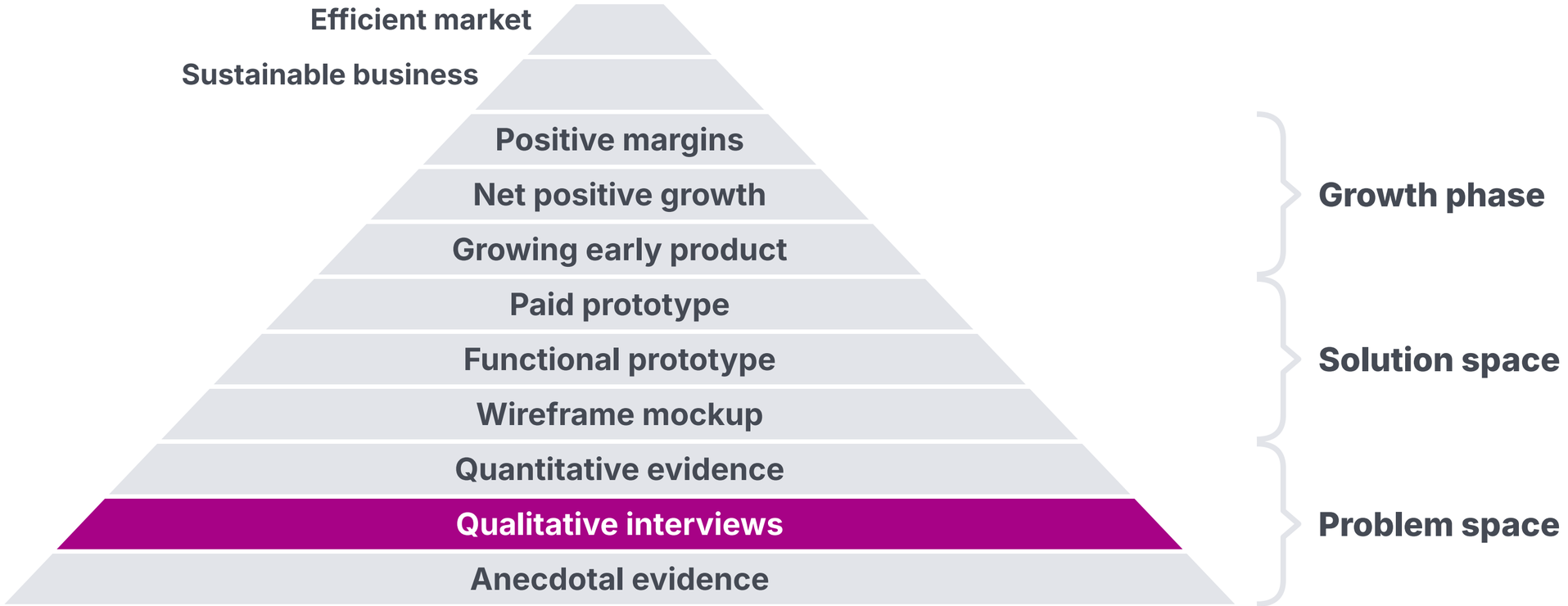
PREVIOUSLY · PROBLEM SPACE (M03)

You know who hurts. Now count who acts



PREVIOUSLY · PROBLEM SPACE (M03)

You stood on the second step



TODAY'S QUESTION · HANDS

What evidence would change your mind?

1:00

CHAPTER 2 OF 9

Your presentations

Three minutes per team to show what your records say since last time.

01 ~~Previously on Ambitious Entrepreneurship~~

02 **Your presentations**

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

YOUR PRESENTATIONS · MAKE IT CREDIBLE

Show us where you're at in 3 minutes or less!

3:00

Facts

Statistics or research

What your segment spends, and where that is from

12 Numbers

Usage or surveys

Conversations held, next steps agreed, spend in € or hours

Quotes

Interviews

The three that decided it

Images

Screenshots or users

The workaround, if they showed it

Verdict: segment, job, stay or switch. Hypothesis: three people likely to pay first

What did you do? **What did you learn?** **How can we help?**

Photos on Pixabay: lil_foot_, 6689062, Surprising_Media, tranmautritam.

INPUT 1 OF 4 · CHAPTER 3 OF 9

Rank it: what your evidence proves

What you know is only as strong as the step it stands on.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 1 OF 4 · RANK IT · THREE QUESTIONS

What is your evidence worth?

1



Which evidence is strongest?

2



When does a survey count?

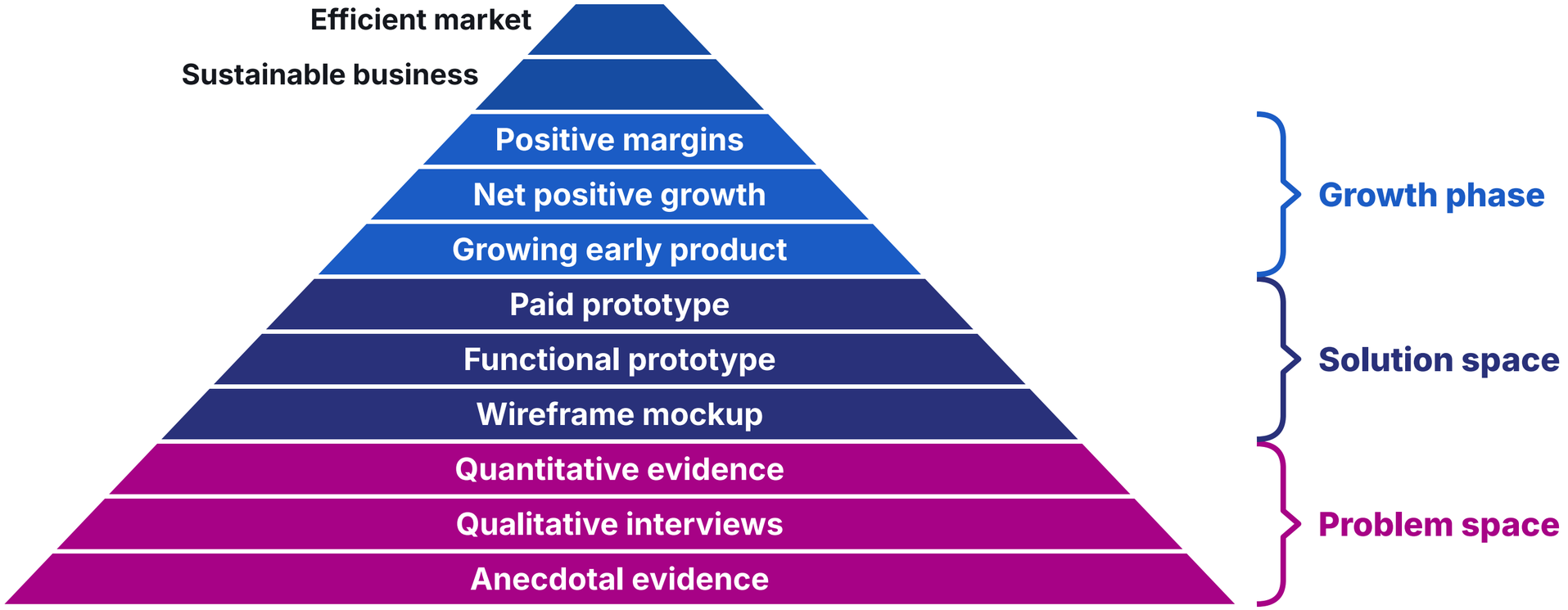
3



What validates a problem?

CORNERSTONE · HIERARCHY OF EVIDENCE

Climb from what they say to what they pay



IN DETAIL

Climb from what they say to what they pay

Anecdotal evidence. One person tells you something, or you lived it yourself. Enough to start looking, never enough to decide.

Qualitative interviews. Conversations about the last time the problem happened. They build your theory: who has the problem, what it costs, what they do today.

Quantitative evidence. Tests how strong the effect is that your interviews suggested: how many in your segment act on it. A survey about past behaviour, or a landing page that counts clicks.

Wireframe mockup. A quick thing that only shows how the solution could look. People react to it, nothing works yet.

Functional prototype. It works. A slice of the product solves the problem for real users, even if you run parts of it by hand.

Paid prototype. Somebody opens their wallet for it. The first rung that counts as validation, everything below is signal.

Growing early product. Customers pay again and new ones arrive. The question moves from whether it works to whether it grows.

Net positive growth. You win more customers than you lose, month after month.

Positive margins. Each sale earns more than it costs to deliver.

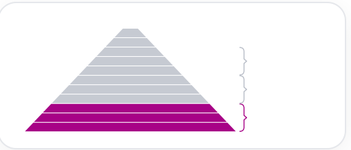
Sustainable business. The business carries itself, without your next funding round.

Efficient market. Others copy you and prices settle. By now everyone has the evidence, and so does every competitor.

Problem space. Rungs 1 to 3. Is the problem real, who has it, and how many?

Solution space. Rungs 4 to 6. Does your solution solve it, and will someone pay?

Growth phase. Rungs 7 to 9. Does it work as a business, again and again?



THE RUNNING EXAMPLE · THE ARM THAT HURTS

Anecdotes diagnose. Interviews theorize. Numbers verify

Anecdote

Under 5 people · subjective



"When I do this, my arm hurts."

Diagnose

Interviews

Under 20 people · unstructured



"Doing this is ok, but that hurts."

Theorize

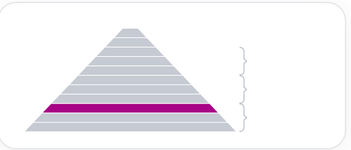
Numbers

Over 20 people · structured



"Sore arm in 90% of runs against a pole."

Verify



SURVEYS · WHEN THEY COUNT

A survey counts only when it asks about the past

FROM

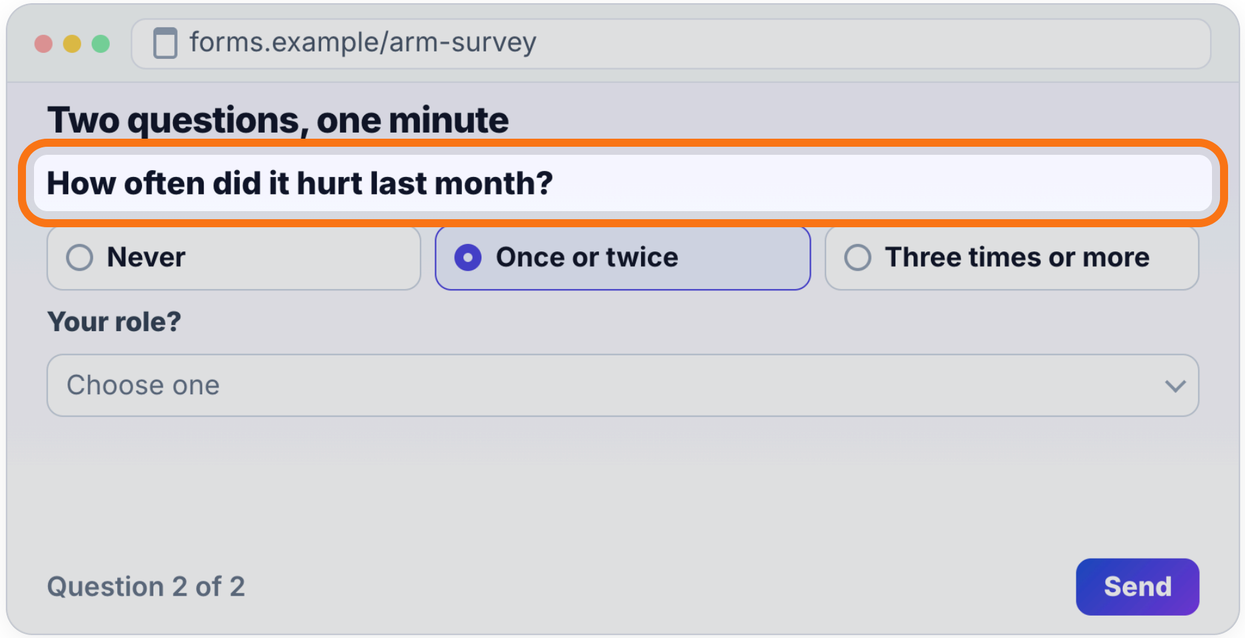
"Tell me about instances when it hurts."

↓

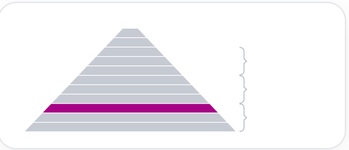
TO

"How often did it hurt last month?"

- **Never mention your solution**
- **Role and context, to count your segment**
- **Short, in their words**

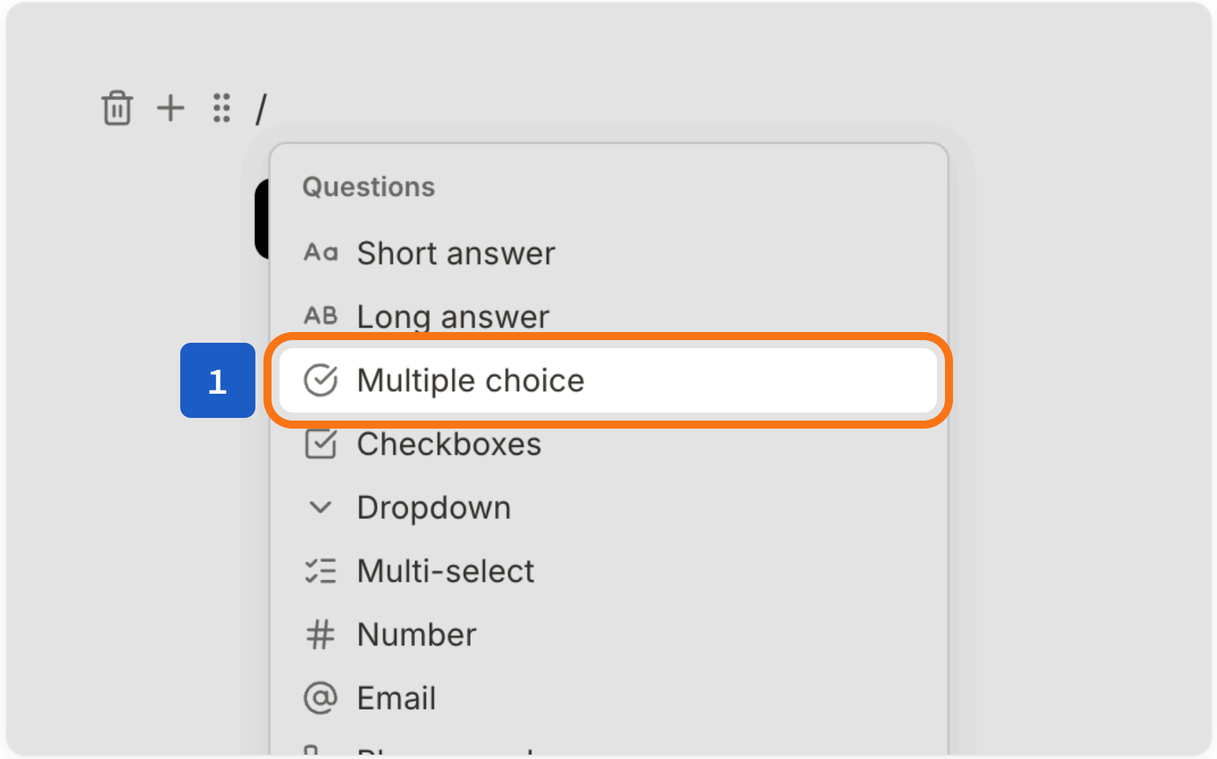


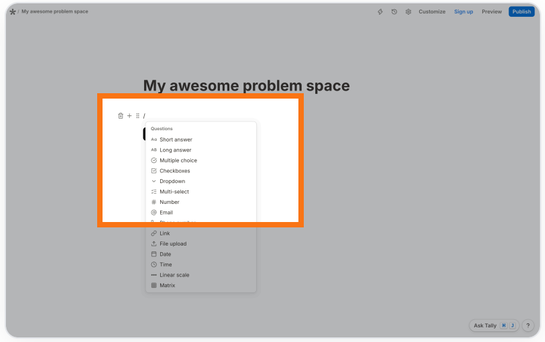
Course rule on surveys, 2026-10-19. Hypothetical bias: Murphy et al. (2005), 28 studies, Schmidt & Bijmolt (2020), 77 studies. Survey prototyped for this course.



HOW-TO · SURVEY · Tally · STEP 1 OF 4

Build the form in ten minutes

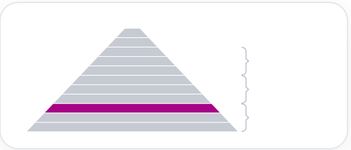




1. **Build the form**
2. Ask about the past
3. Send it to people who acted
4. Let AI count, then check

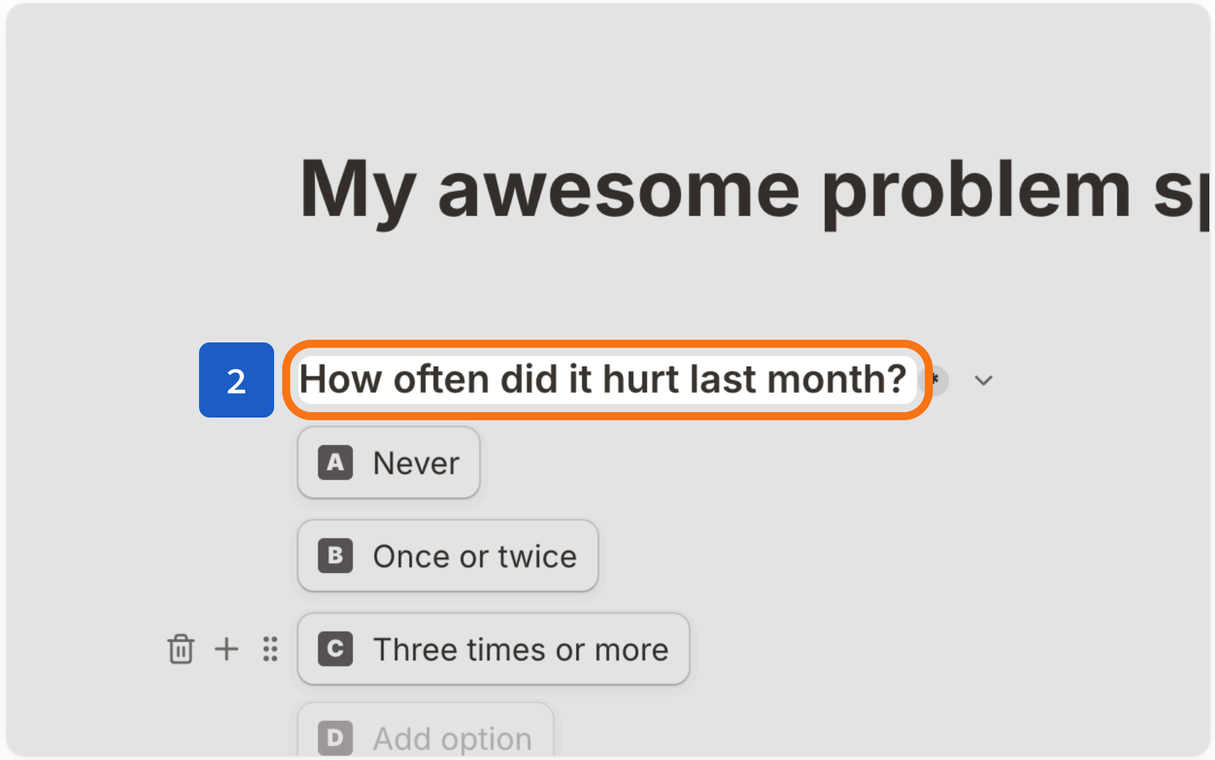
Closed questions only.

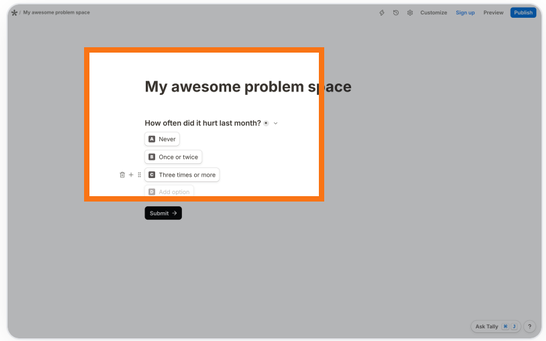
Screenshots of tally.so as of 26 October 2026, captured with scripts/shots.py. Example form typed for this deck. Any form tool works.



HOW-TO · SURVEY · STEP 2 OF 4

Ask about the past

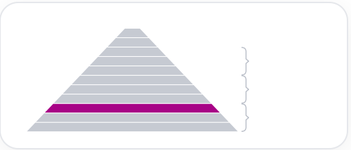




1. Build the form
2. **Ask about the past**
3. Send it to people who acted
4. Let AI count, then check

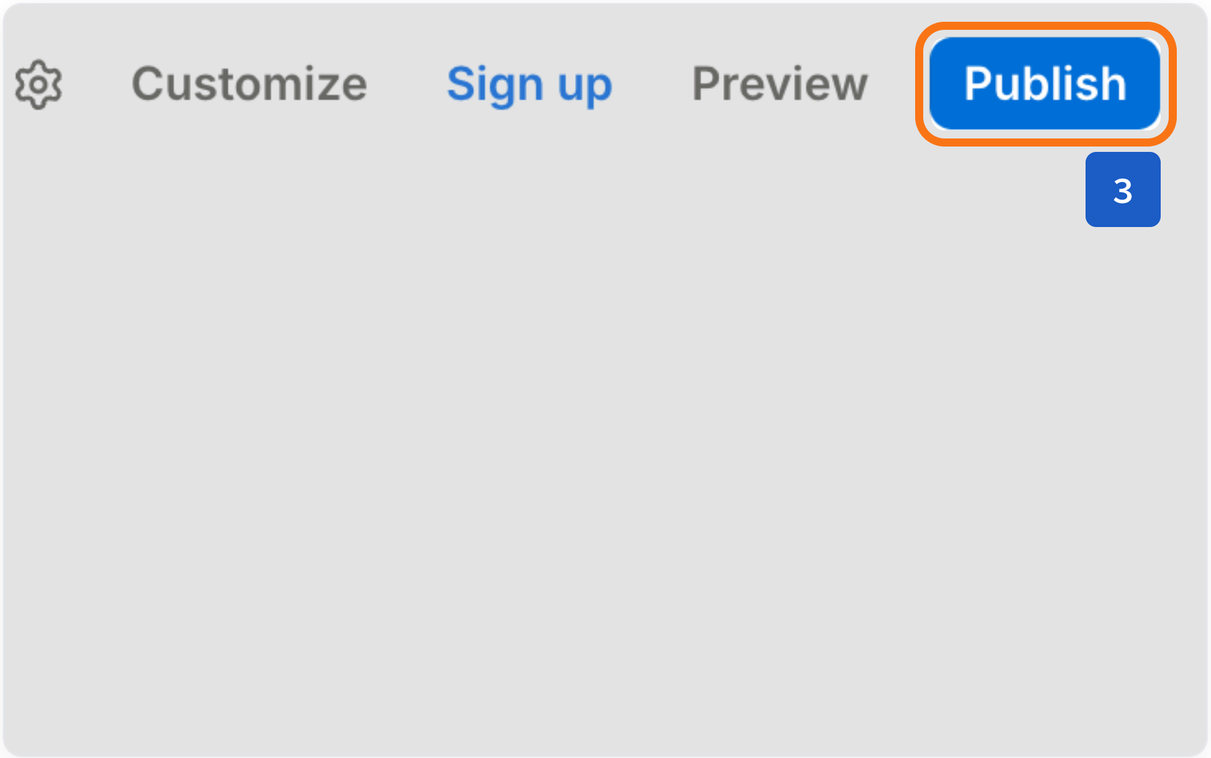
One question about last month, three closed answers

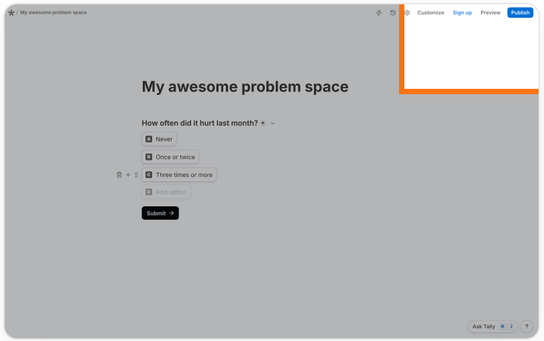
Screenshots of tally.so as of 26 October 2026, captured with scripts/shots.py. Example form typed for this deck. Any form tool works.



HOW-TO · SURVEY · STEP 3 OF 4

Send it to people who acted

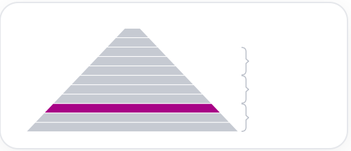




1. Build the form
2. Ask about the past
3. **Send it to people who acted**
4. Let AI count, then check

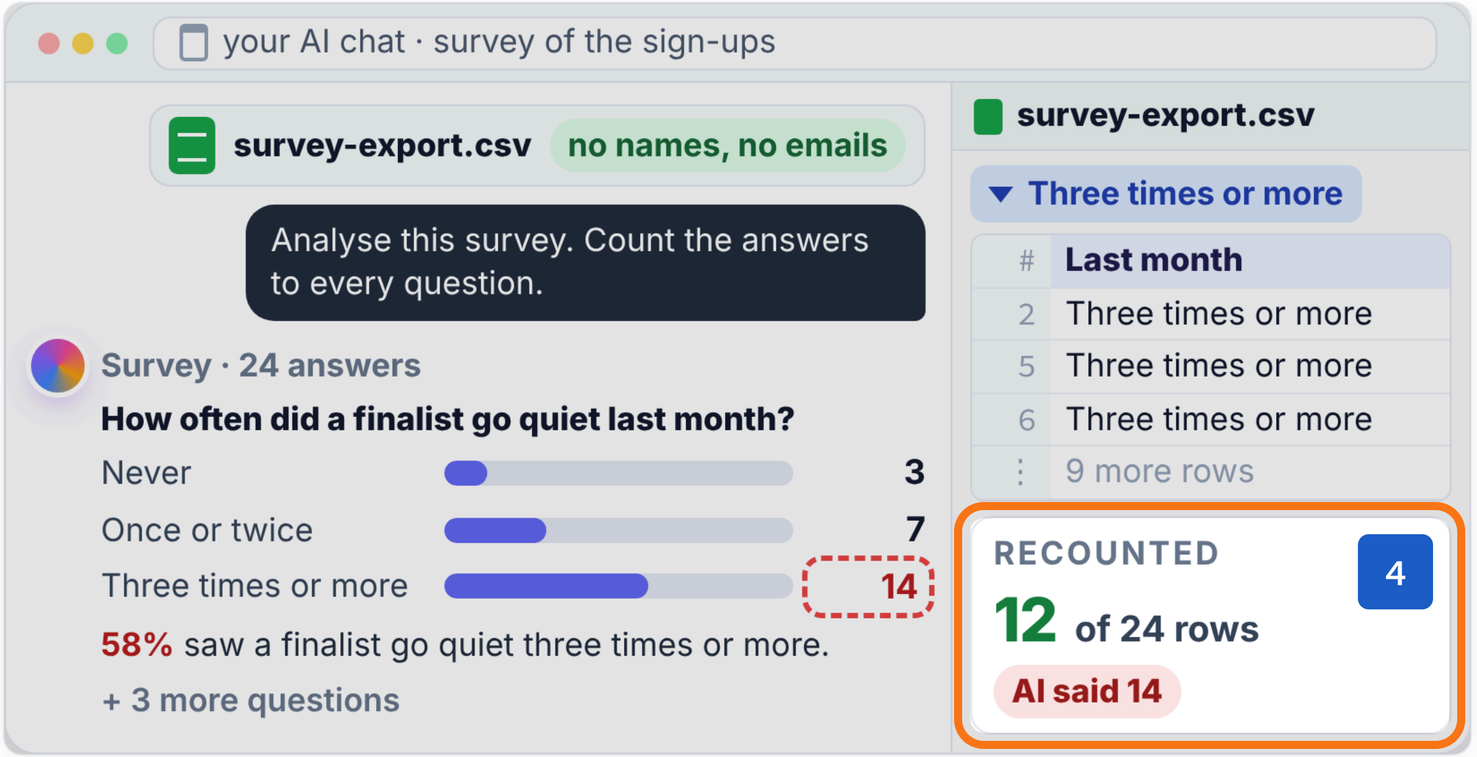
Only to people who already acted, your sign-ups

Screenshots of tally.so as of 26 October 2026, captured with scripts/shots.py. Example form typed for this deck. Any form tool works.



HOW-TO · SURVEY · STEP 4 OF 4

Let AI count. Check every number



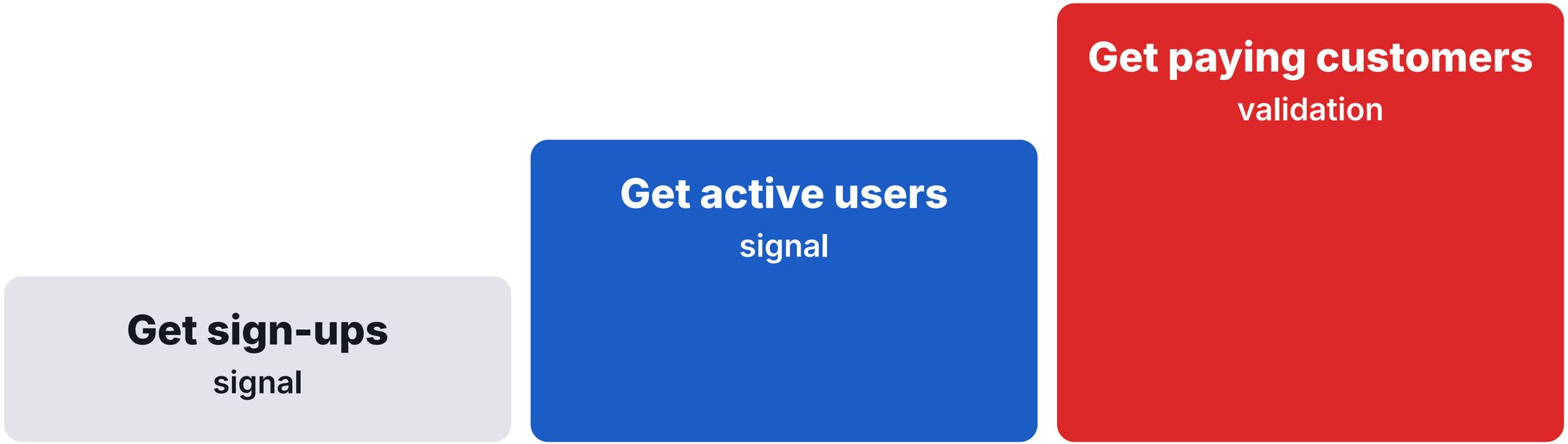
1. Build the form
2. Ask about the past
3. Send it to people who acted
4. **Let AI count, then check**

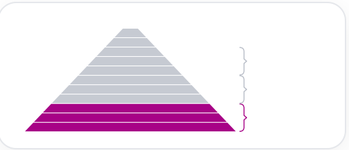
Export without names and emails.

ILLUSTRATIVE Screen prototyped for this course, answers and numbers made up.

MANTRA

Real validation comes through the credit card





NOW YOU · AS A TEAM · 3 MINUTES

Put your evidence on the pyramid

Anecdote What did one person say?

Interviews What did many say?

Number What did 20+ do?

Limit What does it not prove?

3:00

OUTPUT

Your top step, honestly

INPUT 1 OF 4 · RANK IT · THE ANSWERS

Rank it: saying, doing, paying

1



Which evidence is strongest?

What people do and pay beats what they say. Numbers verify

2



When does a survey count?

Only when it asks about the past and never mentions your solution

3



What validates a problem?

Sign-ups and users are signal. Validation comes through the credit card

INPUT 2 OF 4 · CHAPTER 4 OF 9

Draw it: the line before the data

Write down what would change your mind before the first visitor arrives.

01 ~~Previously on Ambitious Entrepreneurship~~
02 ~~Your presentations~~
03 ~~Rank it: what your evidence proves~~
04 **Draw it: the line before the data**
05 **Launch it: a page that tests your problem**
06 **Count it: visitors from your segment**
07 Wrap-up and check-out
08 Workshop: launch a page that tests your problem
09 Case discussion

INPUT 2 OF 4 · DRAW IT · THREE QUESTIONS

What would change your mind?

1



Why write the line before the data?

2

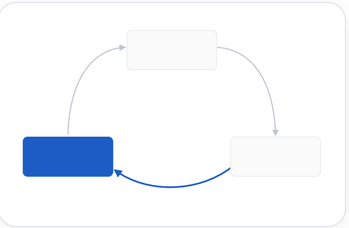


What goes on a test card?

3

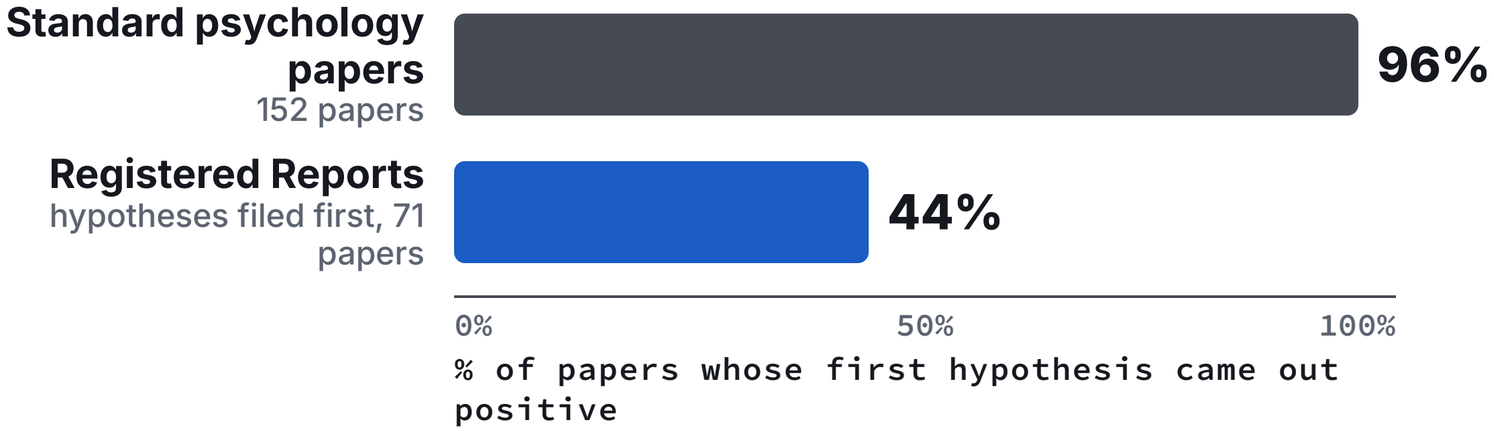


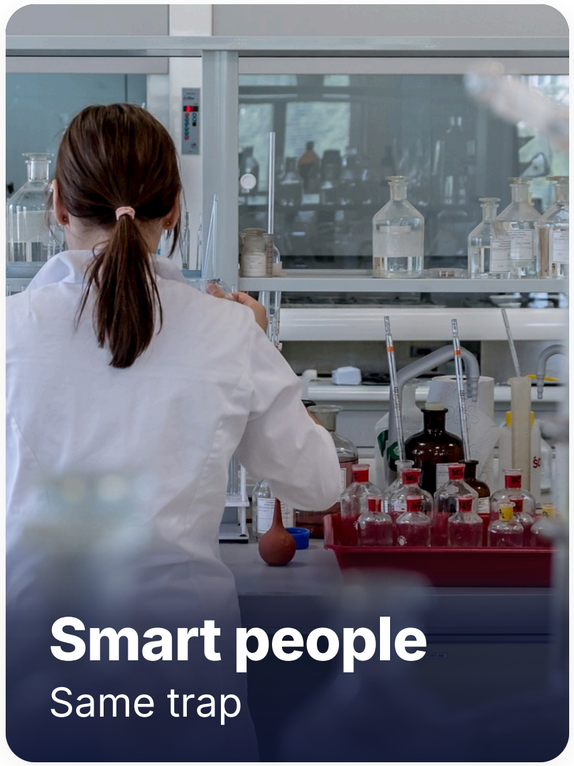
What if you miss the line?



WHY IT MATTERS

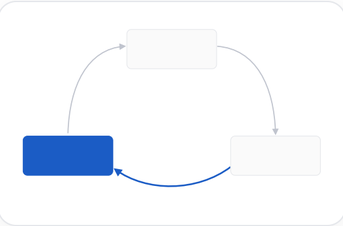
Decide after the data, and everything says yes





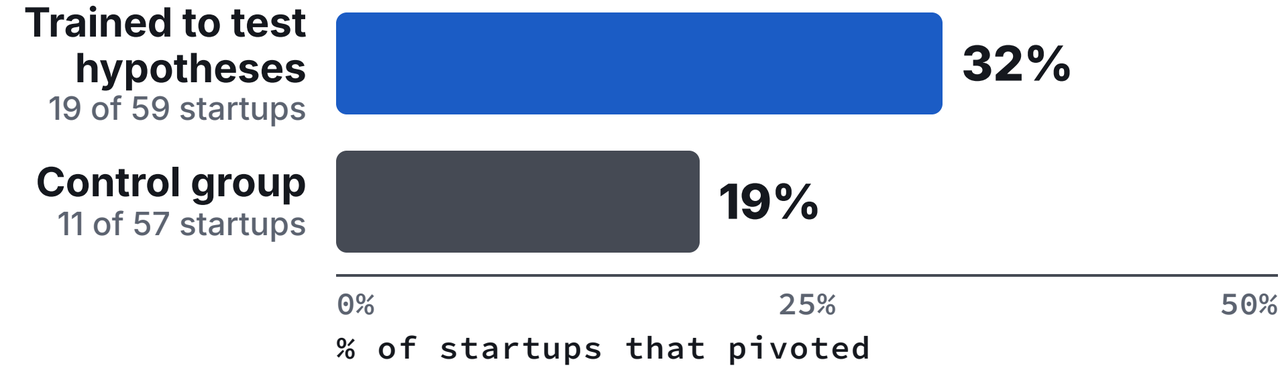
57% → 8%
heart trials with a benefit, before and after 2000

87%
of 67 top-journal trials needed a correction for their outcomes



WHY IT MATTERS · FOUNDERS

Founders taught to test pivot more

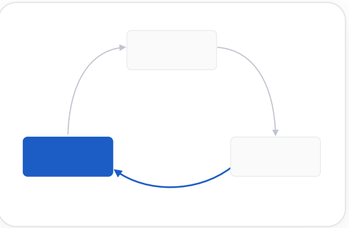




116
Italian startups, randomized

759
firms in 2024: more ideas stopped

RESEARCH Camuffo, Cordova, Gambardella & Spina (2020), *Management Science*. Camuffo et al. (2024), *Strategic Management Journal*. Bars to scale. Photo: StartupStockPhotos on Pixabay.

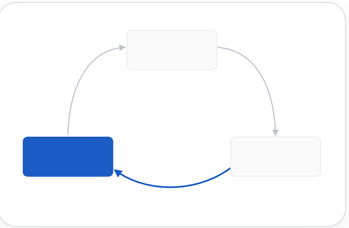


TEST CARD · BEFORE ANYONE VISITS

Write the test before the page exists

We believe	[segment] will [action] when we promise [outcome]
To verify, we	publish the page, bring [N] visitors from [sources]
And measure	the share of segment visitors who take the action
We are right if	[X]% or more by [date], logged today
Then	pass: [next step] · fail: switch segment or problem

FRAMEWORK Test card after Bland, D. J., & Osterwalder, A. (2019). *Testing Business Ideas*. Wiley. Course material, redrawn with a "Then" row.



EXAMPLE · A TEST CARD

Even an interview test gets a number

We believe	new founders struggle to structure their processes
To verify, we	hold 15 interviews with early-stage founders
And measure	their difficulty, rated from 1 to 10
We are right if	10 out of 15 rate it 7 or higher
Time	2 weeks

ILLUSTRATIVE Example test card. Name removed.

WITH YOUR AGENT

Even an interview test gets a number

Your agent `test-card`

Fill a test card from my Pain Map and notes. Leave the line empty, we set it.

`interviews-no-names.txt`

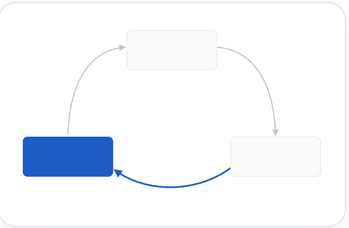
Done. Belief, test and metric come from your 15 interviews. **The line is yours.** How many of 15 must rate it 7 or higher?

10 of 15, in 2 weeks. Log it with today's date.

Logged in the Kill Criteria Log, before the first interview.

Reply...

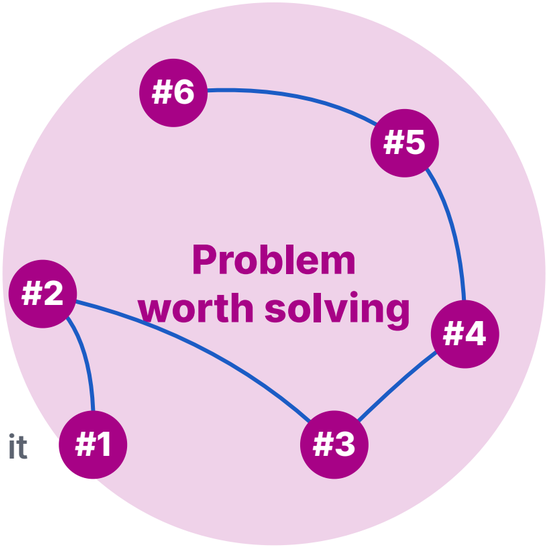
ILLUSTRATIVE Example test card. Name removed.



WHEN YOU MISS THE LINE

Missed the line? Move inside the problem space

You are passionate about it
Your audience is passionate about it
You can make a good living from it

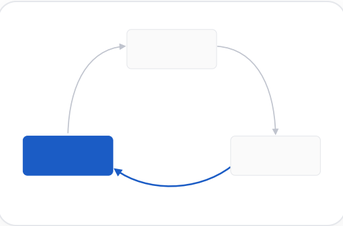


Each hop is the next segment or problem

Here, a small hop inside the problem space
Type 2. Next segment, neighbouring problem

The long jump, leave for another problem space
Type 2 too

Later, type 1
Same problem, another solution



NOW YOU · AS A TEAM · 3 MINUTES

Write your test card

We believe [segment] will [action]

We measure share of segment visitors

We pass at [X]% by [date]

Then pass, and fail

3:00

OUTPUT

One card, dated today

INPUT 2 OF 4 · DRAW IT · THE ANSWERS

Draw it: the line comes first

1



Why write the line before the data?

Decide after the data, and everything says yes. Founders who test pivot more

2



What goes on a test card?

Segment, action, promise, visitors and sources, and the share that passes by a date

3



What if you miss the line?

Move inside the problem space: the next segment or problem

Launch it: a page that tests your problem

A page in their words with one button is your first quantitative test.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 ~~Draw it: the line before the data~~

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 3 OF 4 · LAUNCH IT · THREE QUESTIONS

How does a page test a problem?

1



Why a page before a product?

2

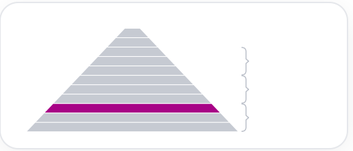


What goes on the page?

3



How fast can you build it?



STORY · **Zappos** · 1999

He sold the shoes before he owned them



1 **No shop has the boots he wants**



2 **He photographs a shop's shoes**



3 **He puts the photos online**

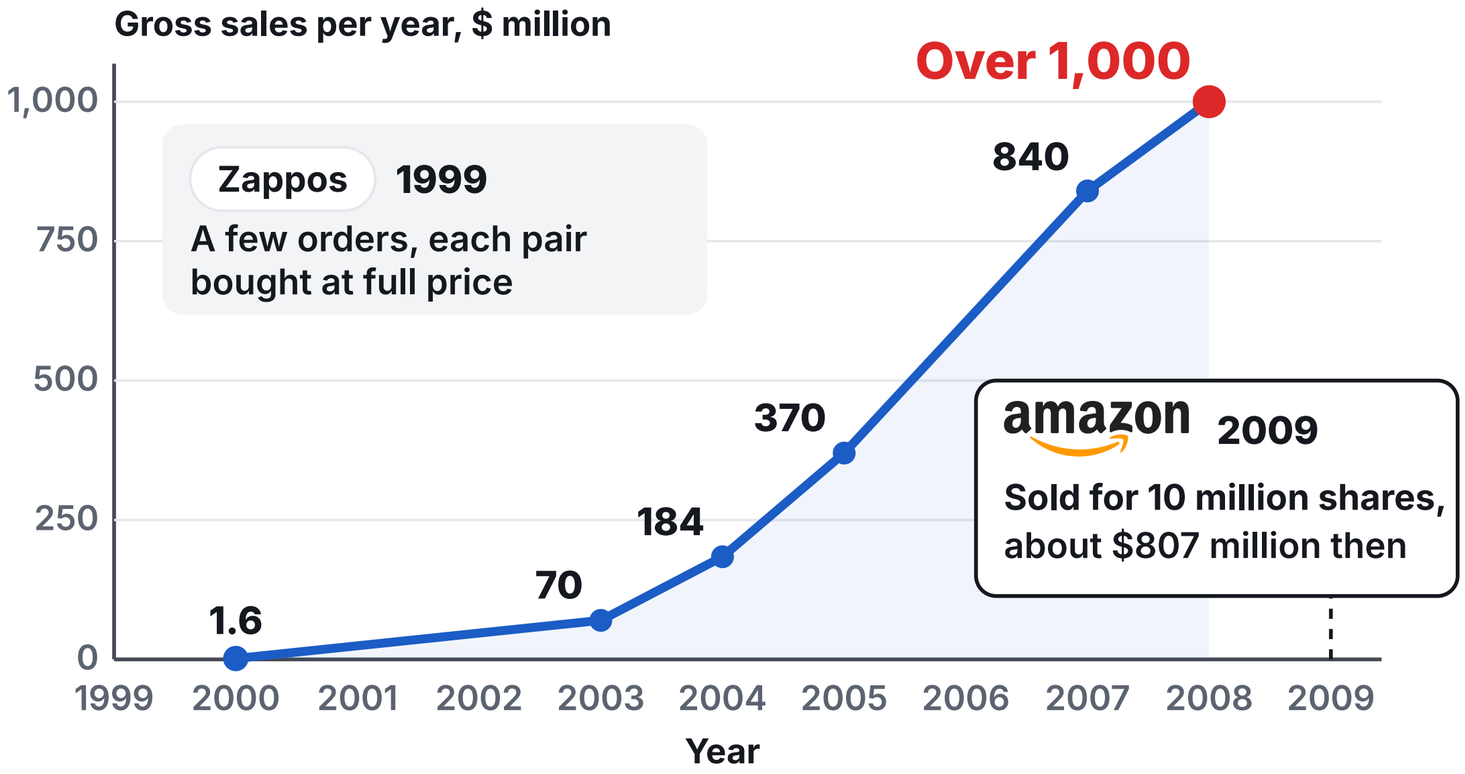


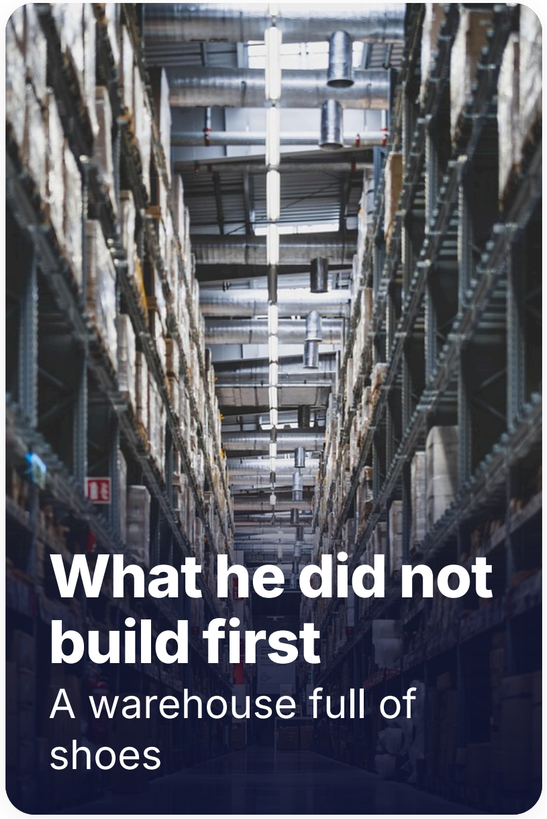
4 **Someone orders. He buys the pair at full price**

DATA Swinmurn, N., as told to D. Eng (2012, 5 September). *Fortune*. Prather, M. (2000, 1 December). Sole proprietor. *Entrepreneur*.

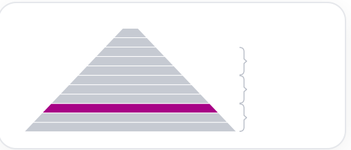
STORY · WHERE IT WENT

From a few orders to $1 billion in nine years





DATA Gross sales from EBSCO Research Starters, Hachette author page. The sale from Amazon press release, 22 July 2009. Logos are trademarks of their owners. Photo: tianya1223 on Pixabay.



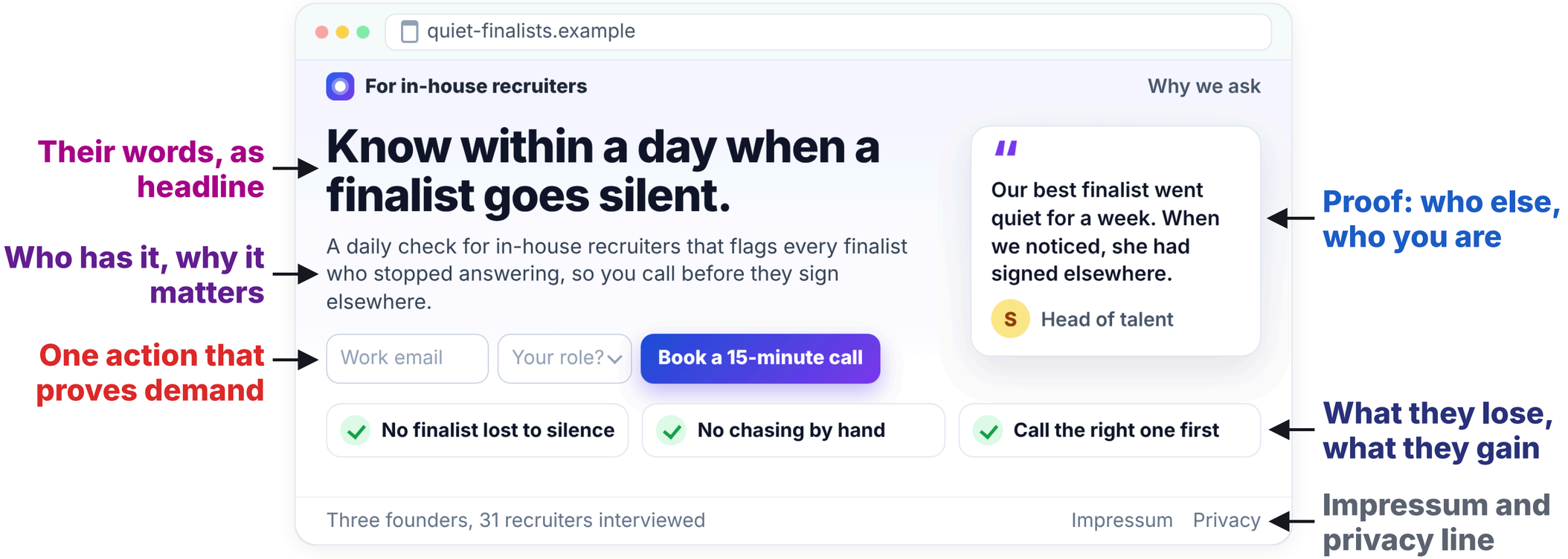
Launch right away. A page is the smallest launch there is.

Only a launch shows whether you meet a need.

YC After Ralston, G. (2017, September 25). YC's essential startup advice. Y Combinator blog. Launching right away is the first piece of his advice.

ANATOMY · PROBLEM-SPACE PAGE

Their problem, their words, one button



ILLUSTRATIVE Page prototyped for this course, the made-up recruiting team of the workshop.

A REAL PAGE · PaperPulse · READ IT TOP DOWN

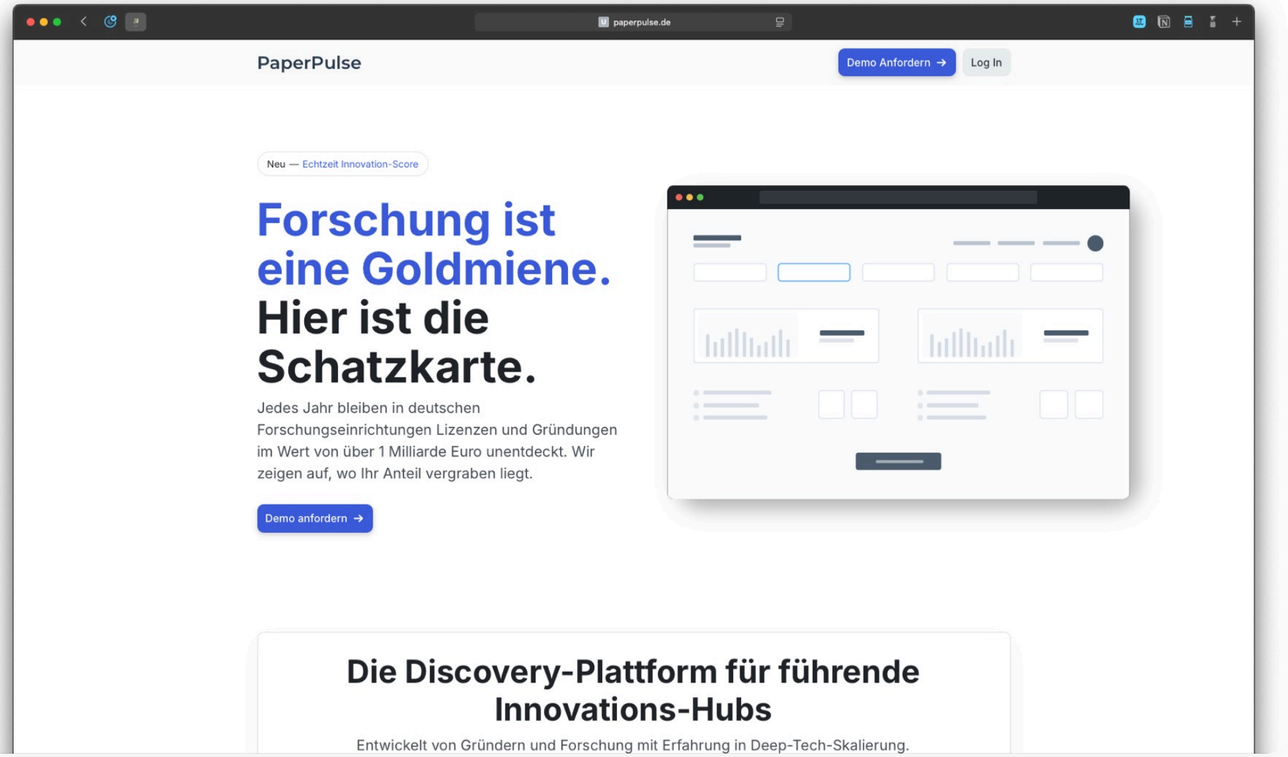
Problem first. The product comes later

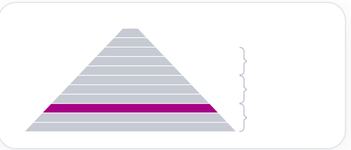
HEADLINE

"Research is a gold mine. Here is the treasure map."

ONE ACTION

Request a demo





LANDING PAGE CONTENT · TIP 1 OF 4

Lead with the outcome they want

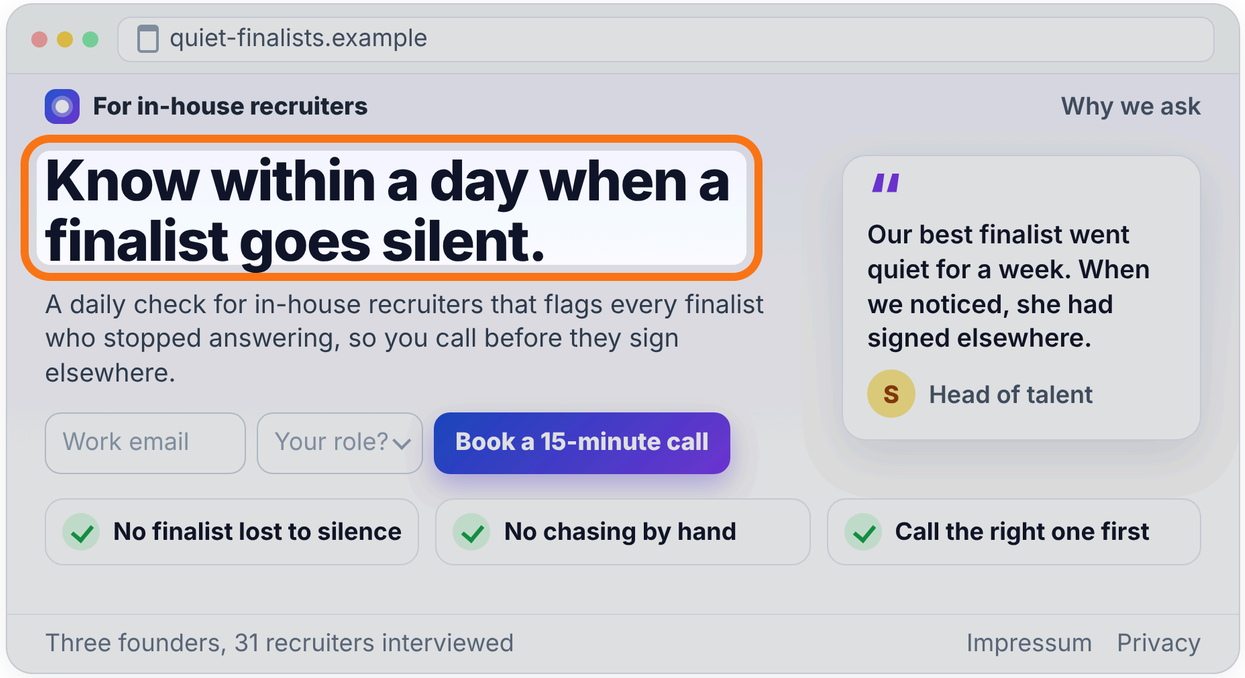
FROM

AI-powered recruiting platform

↓

TO

Know within a day when a finalist goes silent



ILLUSTRATIVE Page prototyped for this course, the made-up recruiting team of the workshop.

LANDING PAGE CONTENT · TIP 2 OF 4

One line says what, for whom, which change

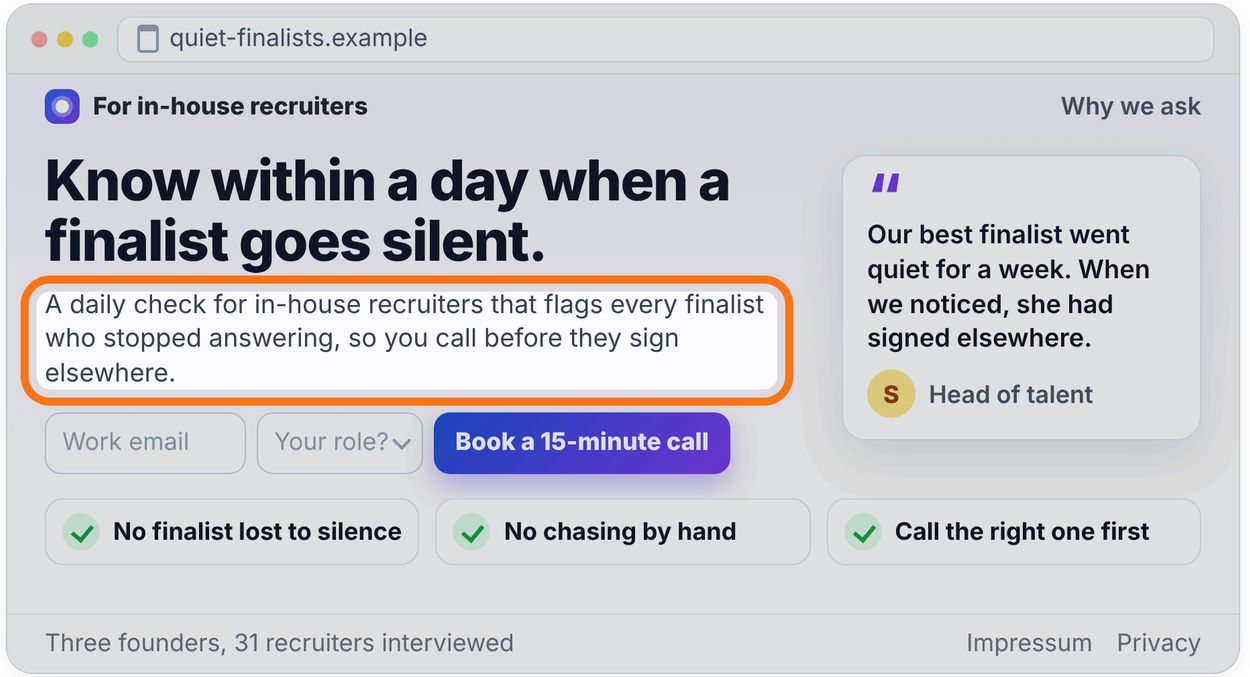
FROM

It's a recruiting tool

↓

TO

A daily check for in-house recruiters that flags every finalist who stopped answering



ILLUSTRATIVE Page prototyped for this course, the made-up recruiting team of the workshop.

LANDING PAGE CONTENT · TIP 3 OF 4

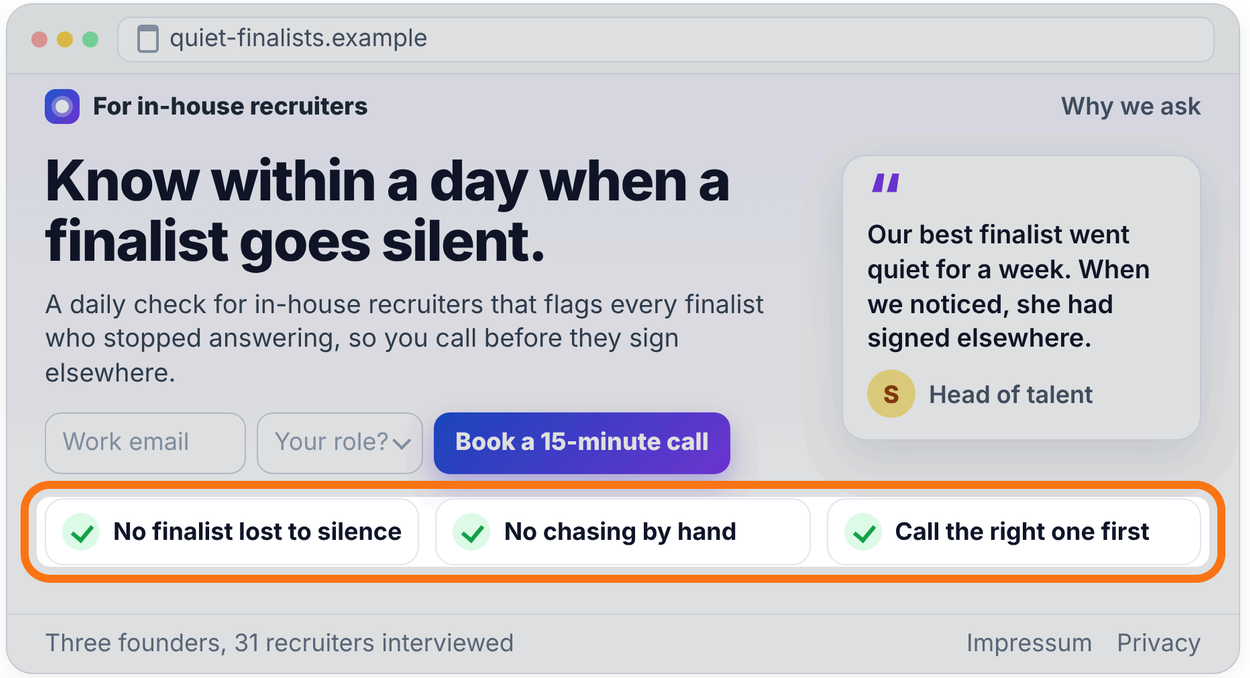
Benefits, not features

FROM

Features of your product

TO

No finalist lost to silence. No chasing by hand.



LANDING PAGE CONTENT · TIP 4 OF 4

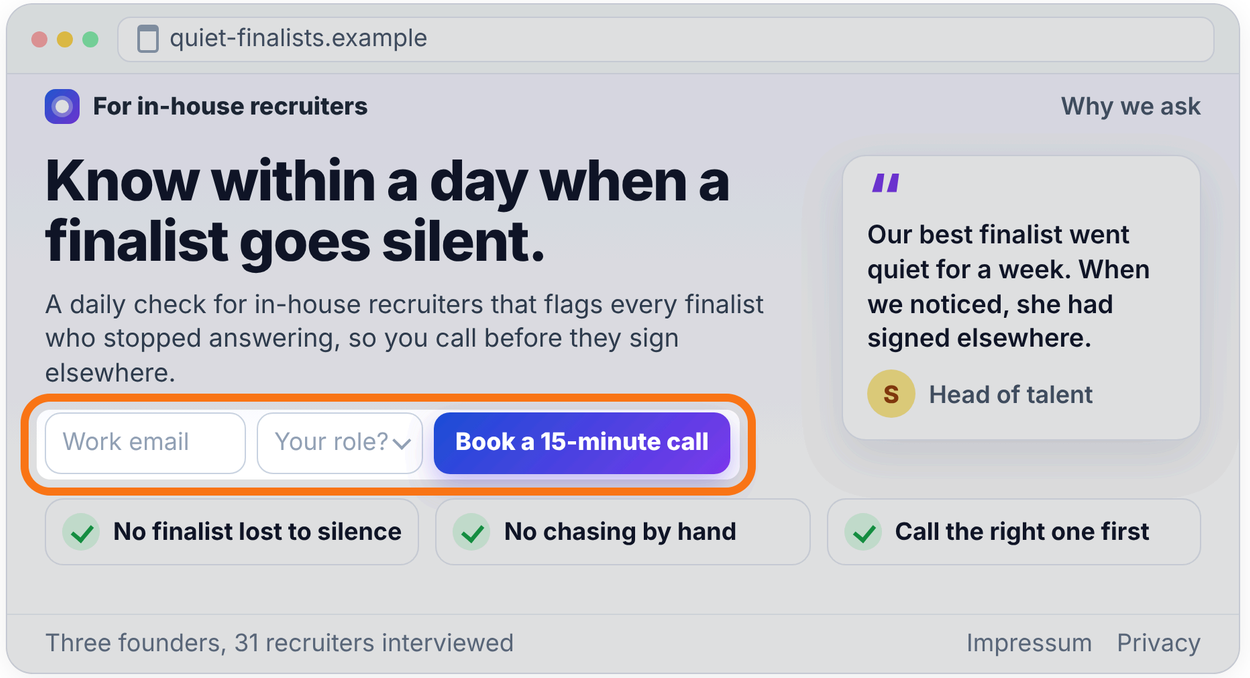
Ask for the action that proves demand

FROM

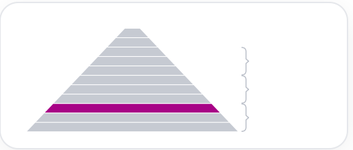
Learn more

TO

Apply now · Book a call · Buy



ILLUSTRATIVE Page prototyped for this course, the made-up recruiting team of the workshop.



VOTE · HANDS

Which button proves the most demand?

1. **Follow us**
2. **Join the waitlist**
3. **Book a 15-minute call**
4. **Pre-order**

1:00

AI COPYWRITING

Don't write from scratch. Start from their words

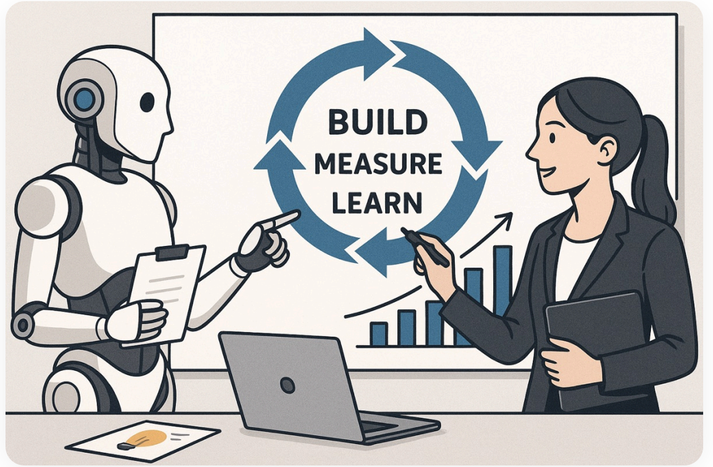
Your agent `hero-copy`

Act as a senior copywriter. Based on these interview notes [PASTE NOTES], write 5 variations of a Landing Page Hero Section (Headline + Subheadline + CTA) that addresses the core pain point directly.

`notes-no-names.txt`

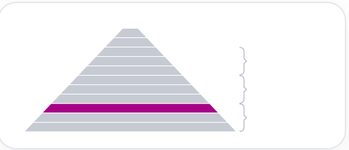
5 hero sections. Each one quotes a line from your notes.

Reply...



A starting point, not the final copy.

ILLUSTRATIVE Chat prototyped for this course, the reply made up.



NOW YOU · ALONE · 3 MINUTES

Your headline, from one quote

Quote What did they say, word for word?

Problem In their words

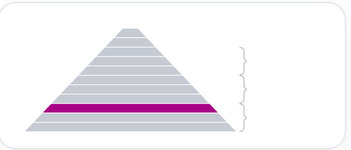
Outcome What changes for them?

Button One action

3:00

OUTPUT

One hero section



TOOLING TO TEST A PROBLEM SPACE

Any builder works. Ship today

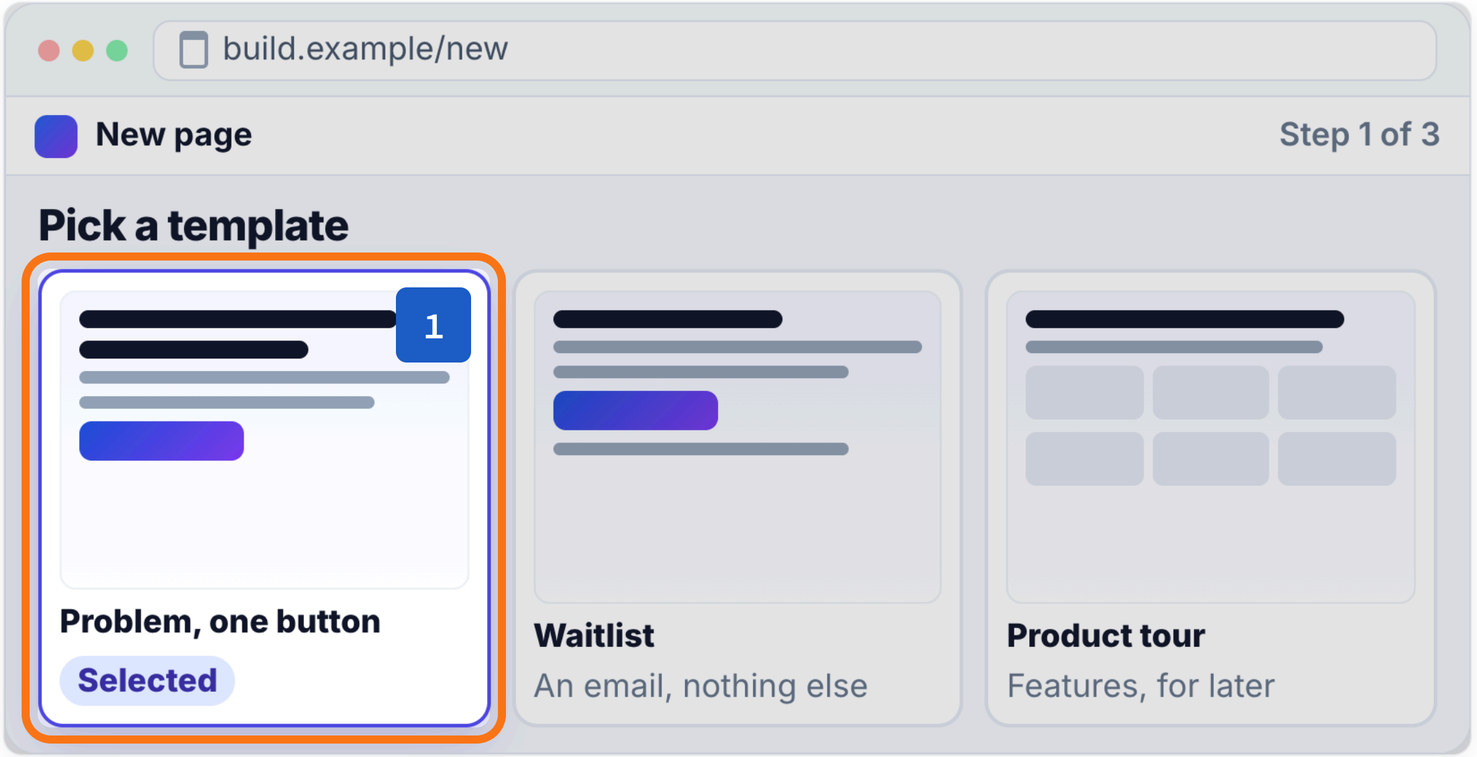
Fast and simple → **Flexible**

Notion	Framer	Webflow	Lovable
A doc as a page	Free plan on a Framer domain	Free starter, more design	Vibe-coded, free daily credits

Vendor pricing pages, as of October 2026. Logos are trademarks of their owners.

HOW-TO · BUILD A PAGE · STEP 1 OF 4

Pick a template. Skip the design debate



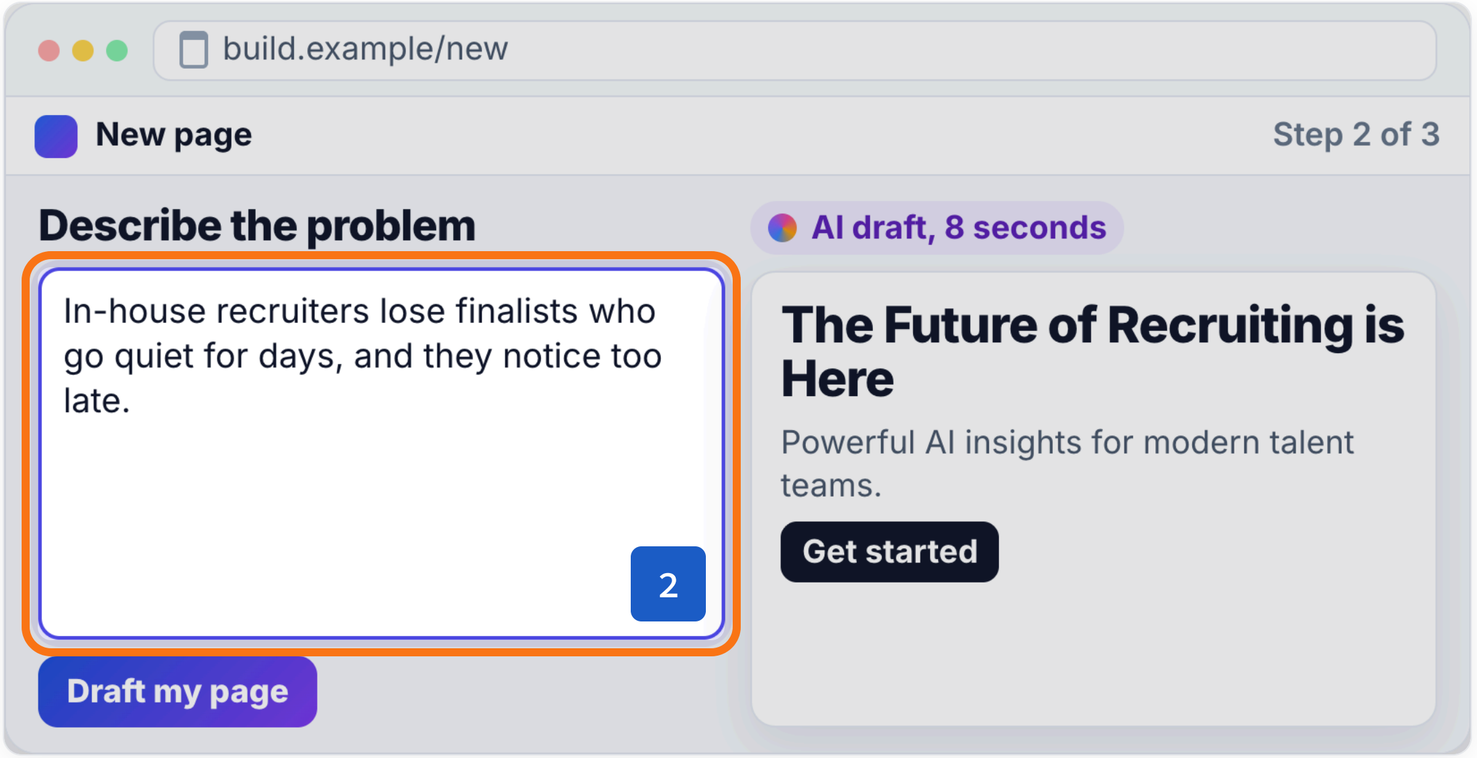
1. **Pick a template**
2. Describe the problem, let AI draft
3. Rewrite in their words
4. Publish on a free subdomain

A hero and one button will do.

ILLUSTRATIVE Screen prototyped for this course, no real product. The made-up recruiting team of the workshop.

HOW-TO · BUILD A PAGE · STEP 2 OF 4

Describe the problem. Let AI draft



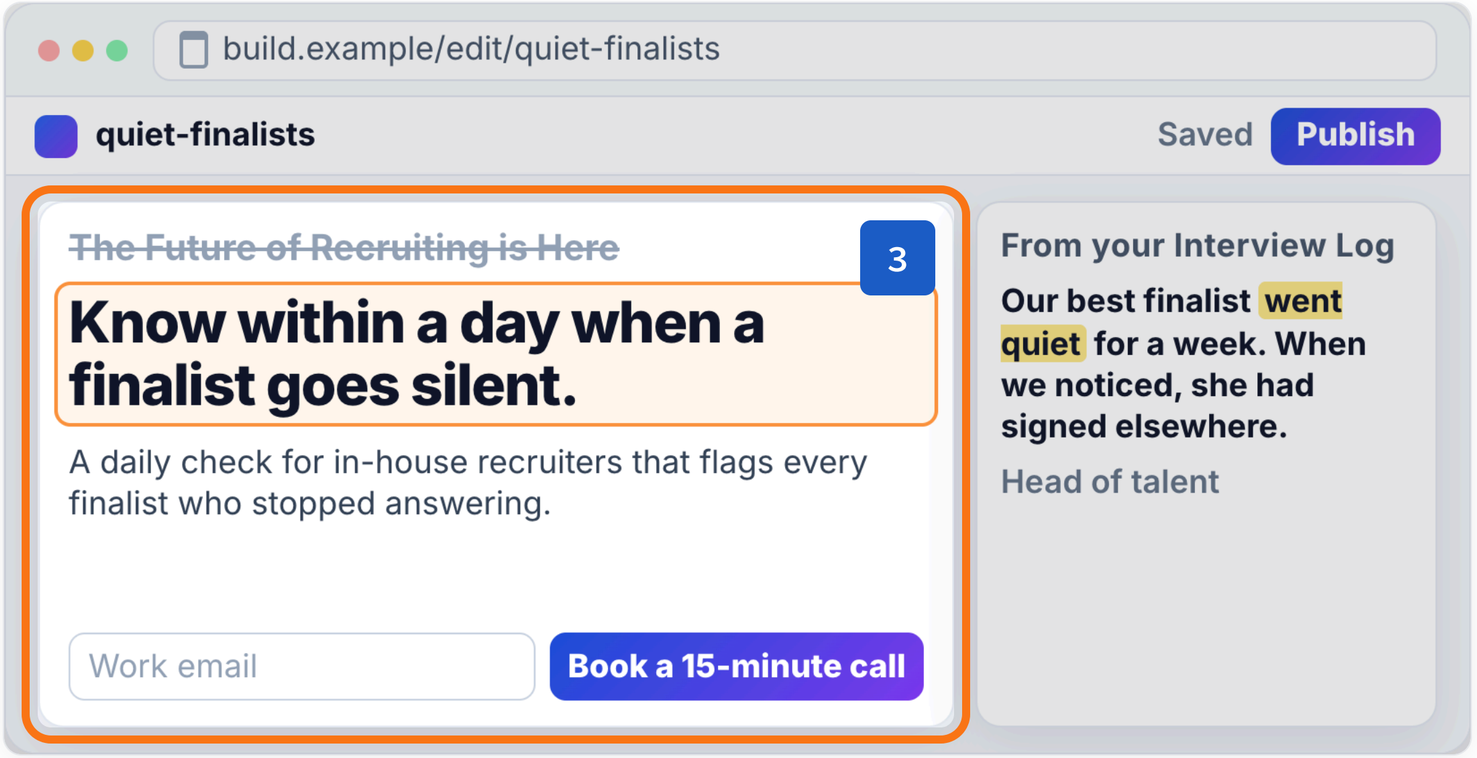
1. Pick a template
2. **Describe the problem, let AI draft**
3. Rewrite in their words
4. Publish on a free subdomain

AI drafts. You decide.

ILLUSTRATIVE Screen prototyped for this course, no real product. The made-up recruiting team of the workshop.

HOW-TO · BUILD A PAGE · STEP 3 OF 4

Rewrite the hero in their words



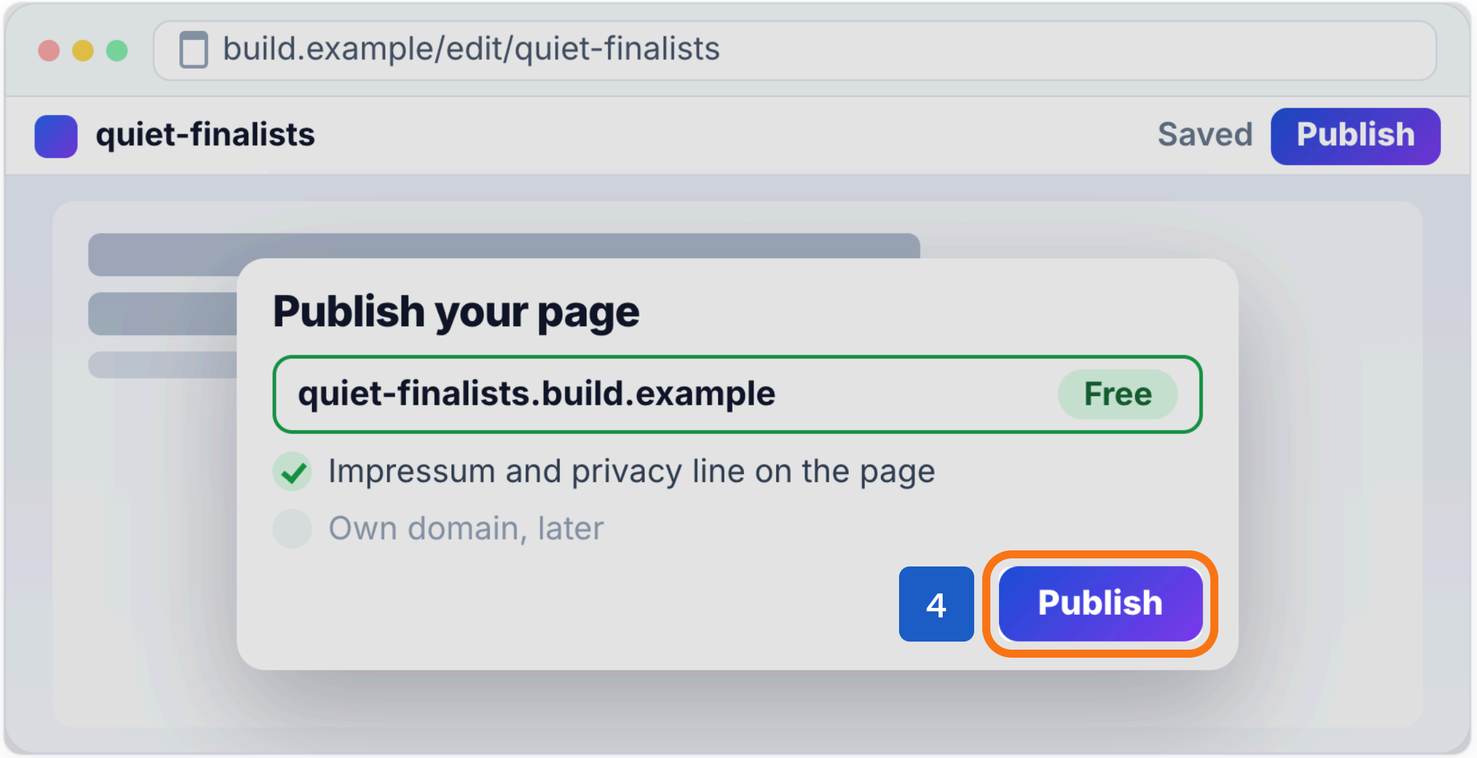
1. Pick a template
2. Describe the problem, let AI draft
3. **Rewrite in their words**
4. Publish on a free subdomain

Their sentence goes here.

ILLUSTRATIVE Screen prototyped for this course, no real product. The made-up recruiting team of the workshop.

HOW-TO · BUILD A PAGE · STEP 4 OF 4

Publish on a free subdomain



1. Pick a template
2. Describe the problem, let AI draft
3. Rewrite in their words
4. **Publish on a free subdomain**

Domains wait until the page earns one

ILLUSTRATIVE Screen prototyped for this course, no real product. The made-up recruiting team of the workshop.

INPUT 3 OF 4 · LAUNCH IT · THE ANSWERS

Launch it: their words, one button

1



Why a page before a product?

Sell before you own it, as Zappos did. Launch right away

2



What goes on the page?

Their problem in their words, the outcome they want, one button that proves demand

3



How fast can you build it?

Today. Any builder works. AI drafts, you rewrite the hero in their words

INPUT 4 OF 4 · CHAPTER 6 OF 9

Count it: visitors from your segment

Bring the people who have the problem, and count only them.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 ~~Draw it: the line before the data~~

05 ~~Launch it: a page that tests your problem~~

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 Case discussion

INPUT 4 OF 4 · COUNT IT · THREE QUESTIONS

Who comes, and who counts?

1



How many visitors do you need?

2

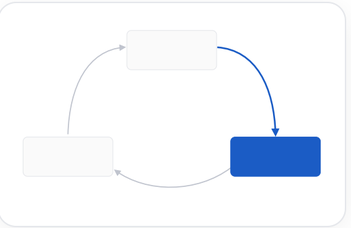


Where do you find them?

3

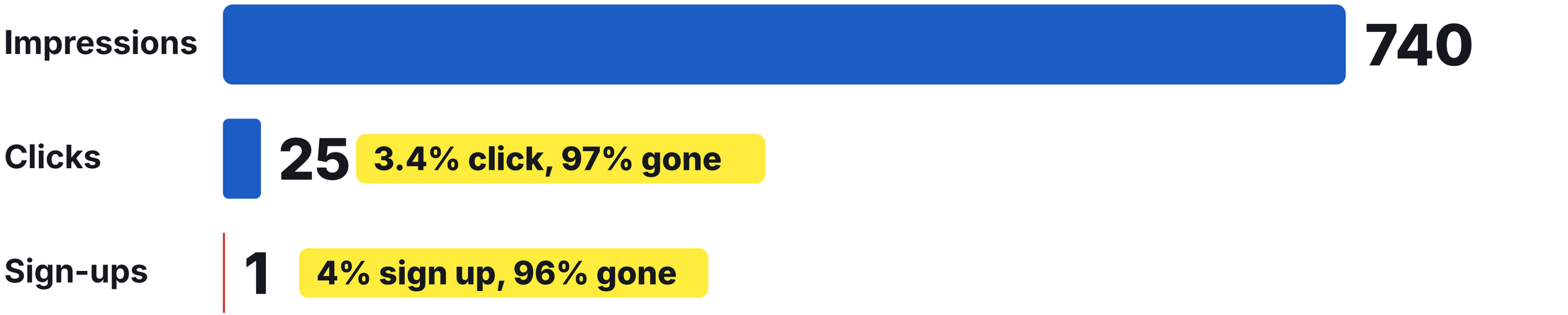


How do you read the result?



LET'S TALK CONVERSION

Work backwards from the sign-ups you need



Work backwards: how many impressions for the sign-ups you need?

REALITY CHECK · DRAWN TO SCALE

One customer takes 9,400 impressions

9,400

impressions for one paying customer

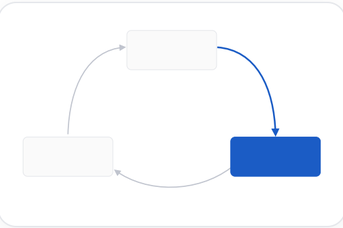
3.4% **click**
Search ads 6.6%, Meta 1.2%

4% **sign up**
SaaS pages 3.8%

8% **pay**
Good is 3 to 5% freemium, 8 to 12% trial



ILLUSTRATIVE The course's example funnel, area to scale. Real rates beside it from LocaliQ 2026, Gupta Media, December 2025, Unbounce 2024, Poyar & Rachitsky 2023, survey of 1,000+ products. Photo: Pexels on Pixabay.



REALITY CHECK · A SIGN-UP IS ONLY A SIGNAL

From sign-up to paying, you lose most again

59 of 64 never pay



5 pay

WHERE IT COMES BACK

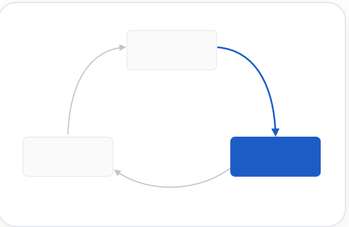
Solution space (M05)
Your sign-ups test your first demo

Finding first users (M08)
You ask them to pay, and count to the payment

Product management (M10)
Sign-ups are no North Star

Why 100 visitors each? 3 paying users need about 40 sign-ups, and those about 400 visitors.

ILLUSTRATIVE The course's example, 5 of 64. Poyar & Rachitsky (2023), survey of 1,000+ products, 3 to 5% is good for self-serve freemium, 8 to 12% for free trials. Arithmetic at 8% and your 10% line, not a forecast.



SPEED MATTERS

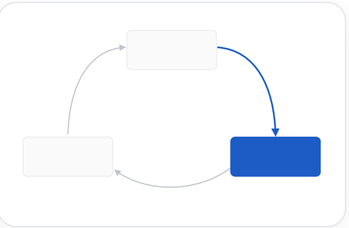
To get, every day, each of you, at least

15 visitors

```
  300 see your link
× 5%  click → 15
  × 7 days → 105
------------------
× 10% sign up → 10
```

300 a day see your link!

ILLUSTRATIVE Arithmetic, redone per student from the homework bar (100 visitors per student) and the default line (10%). The 5% click-through is an assumption, not a benchmark.



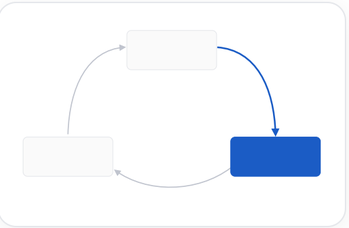
DISTRIBUTION

Go where your segment already is

1% inspiration ~~99% perspiration~~ **99% distribution**

- **B2B**
  - LinkedIn
  - Industry groups
- **Developer tools**
  - Hacker News
  - Product Hunt
- **B2C**
  - Instagram
  - WhatsApp
  - X
- **Niche audience**
  - Discord
  - Slack
- **Niche problem**
  - Facebook groups
  - Forums

Channels by segment type. Logos are trademarks of their owners.

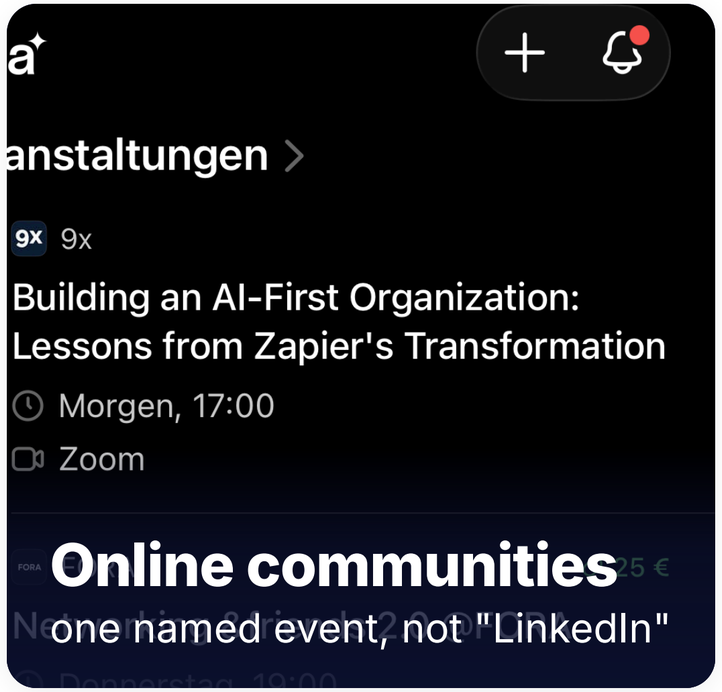


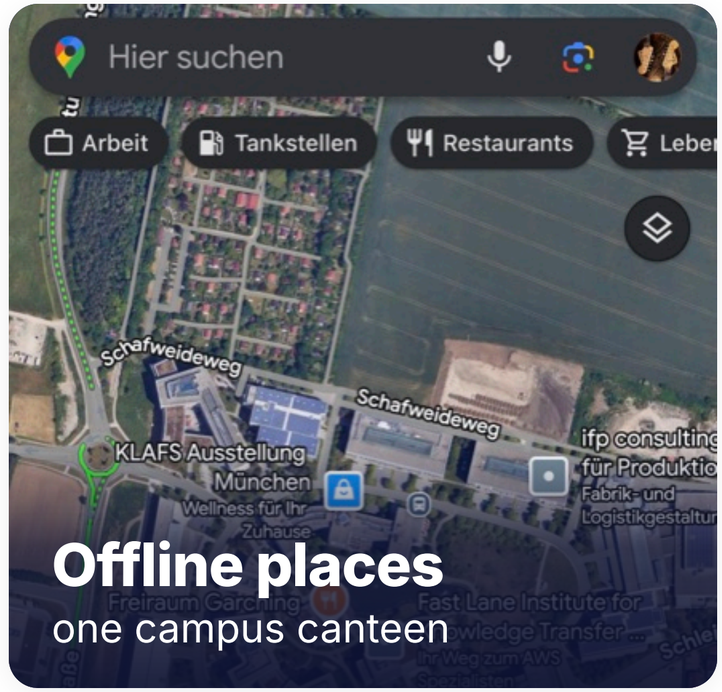
FINDING THE ROOM

Name the room, not the channel

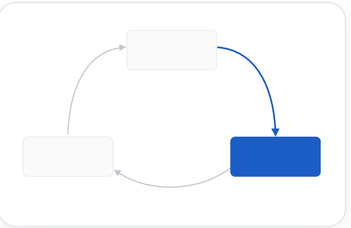
Go where they already complain in public







Phone screenshots, Bastian, 2025. Photo: webandi on Pixabay.



BUILDING AN AUDIENCE

Post what you learn. Talk to who reacts

FROM

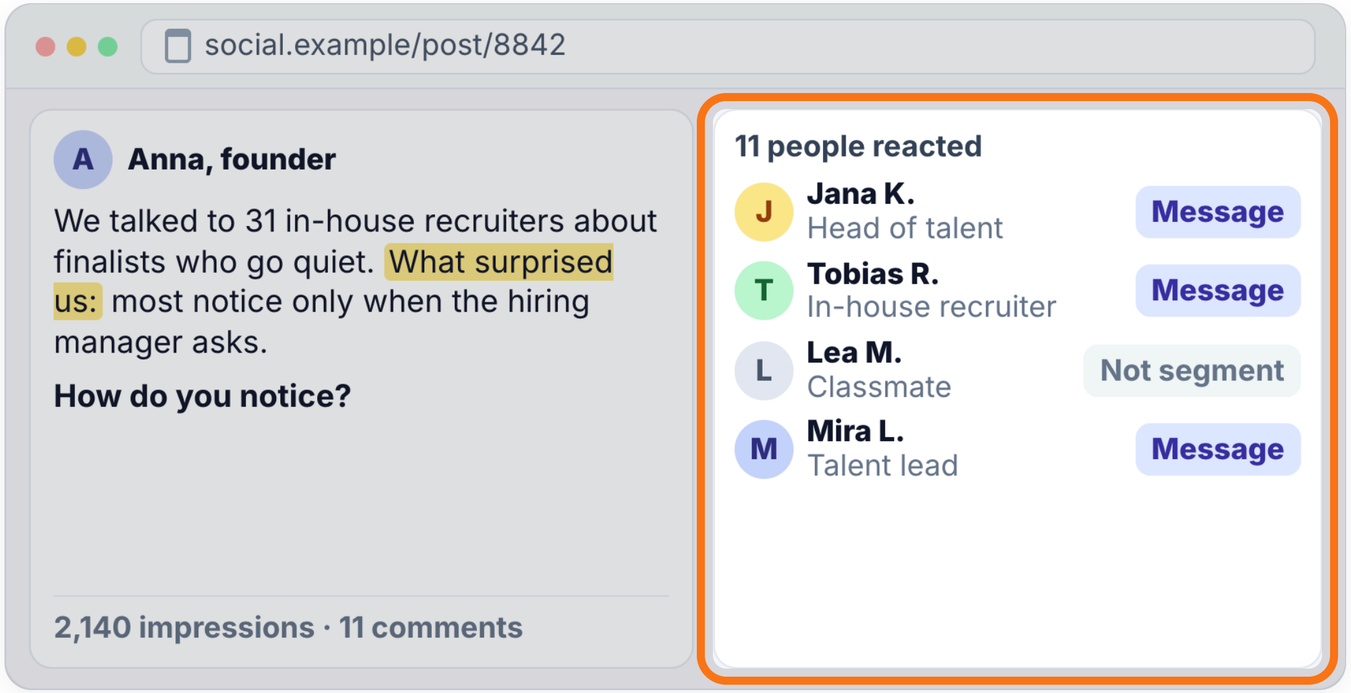
Look what we are building

↓

TO

What you found, what surprised you

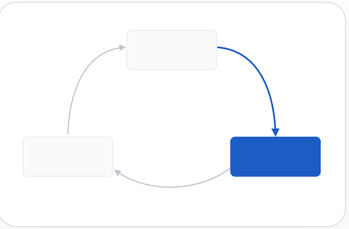
Message everyone who reacts.
Check: are they your segment?



ILLUSTRATIVE Screen prototyped for this course, no real network. Post and people made up, the recruiting team of the workshop.

HOW-TO · LET AI DO THE HEAVY LIFTING

Let an agent search for you, every Monday



your research agent · scheduled tasks

Where in-house recruiters talk

2 Every Monday · 7:00

1 **Task** Find forums, groups and newsletters where in-house recruiters talk about finalists who go quiet. Only new ones, each with a person to contact.

Monday 7:04 · run 6 · 3 new leads, 11 sources

Forum thread **"Finalist ghosted us after the offer call"**, 63 replies
Jana K. Head of Talent, started it
3 ✓ opened

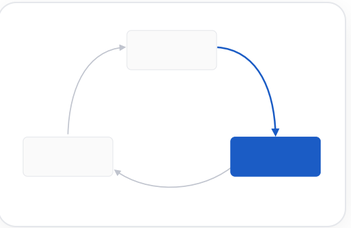
Group for talent acquisition leads, 2,400 members
Tobias R. runs the group
Source ↗

Weekly newsletter for in-house recruiters
Mira L. editor
Source ↗

1. **Ask where your segment talks**
2. **Schedule it weekly**
3. **Open every source**

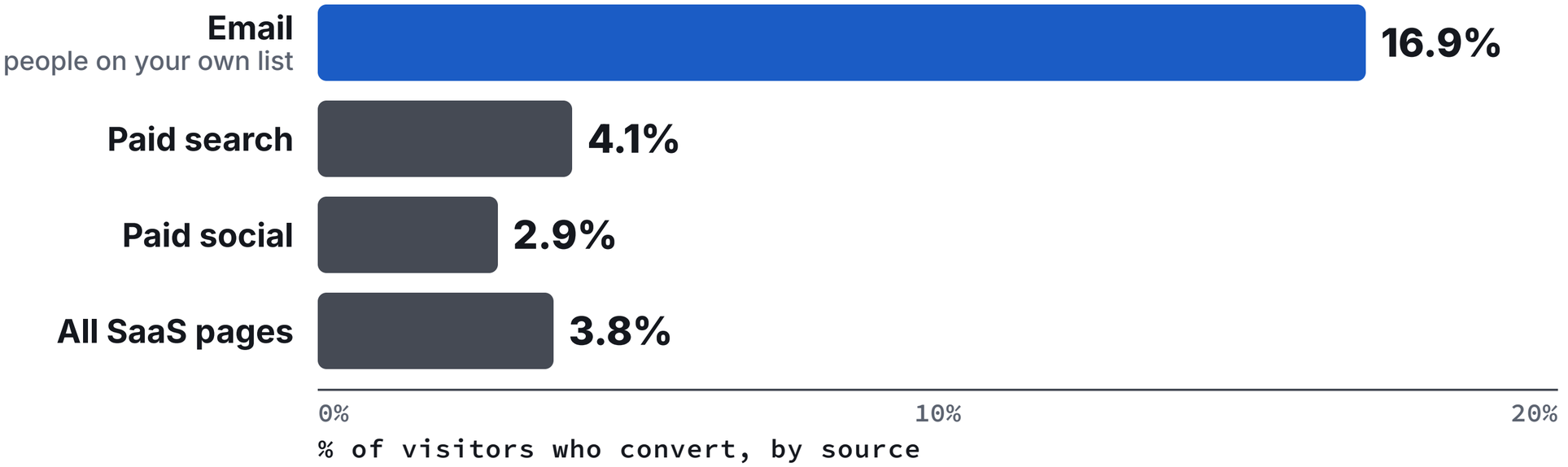
Leads end in a named person.

ILLUSTRATIVE Screen prototyped for this course, forum, group, newsletter and people made up.

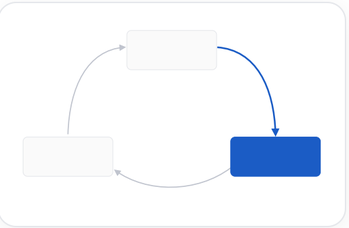


BENCHMARK · WARM AND COLD

Warm traffic converts. Cold traffic costs



DATA Unbounce, Conversion Benchmark Report 2024, SaaS medians, July 2023 to July 2024. A conversion is any click or form event. Bars to scale.



COLD TRAFFIC · BASTIAN'S OWN ACCOUNT · Google Ads

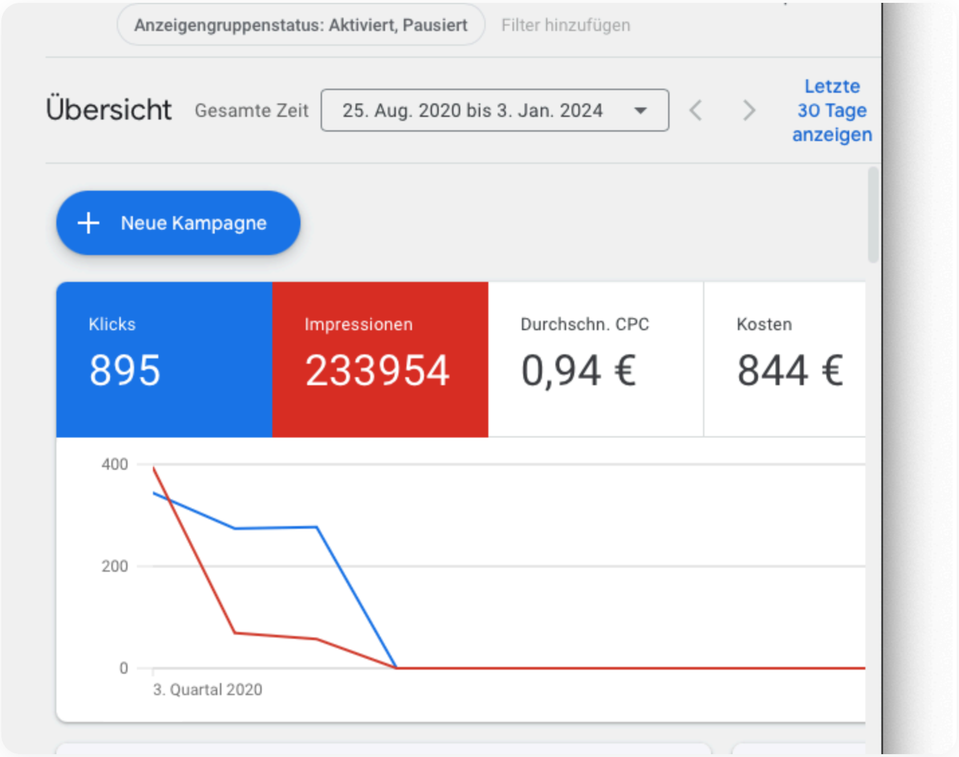
Cold clicks cost €0.94 each

Impressions
Times your ad was shown
233,954

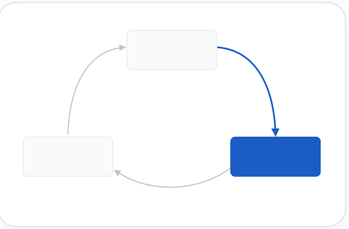
Clicks
Times someone clicked it, 0.38%
895

Spent
What Google charged you
€844

CPC, cost per click
Spent ÷ clicks, €844 ÷ 895
€0.94



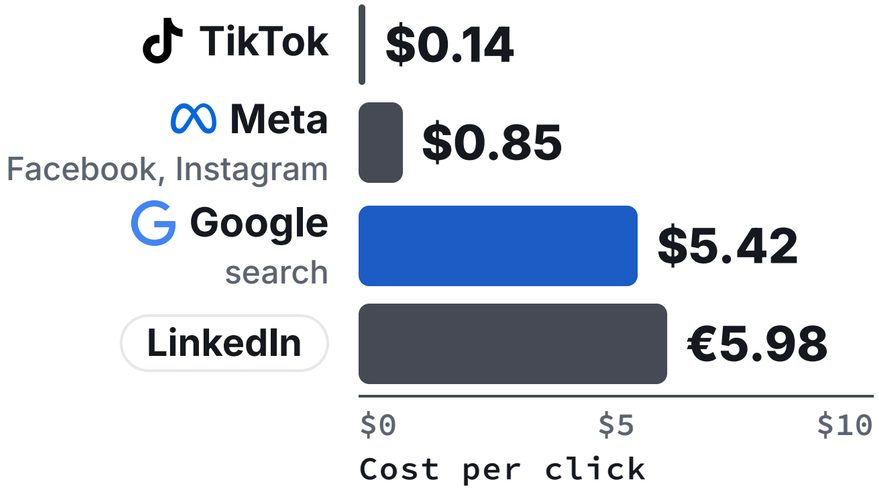
Screenshot of Bastian's Google Ads account, 25 August 2020 to 3 January 2024, clicks in 2020. Click rate 895 ÷ 233,954. Logos are trademarks of their owners.



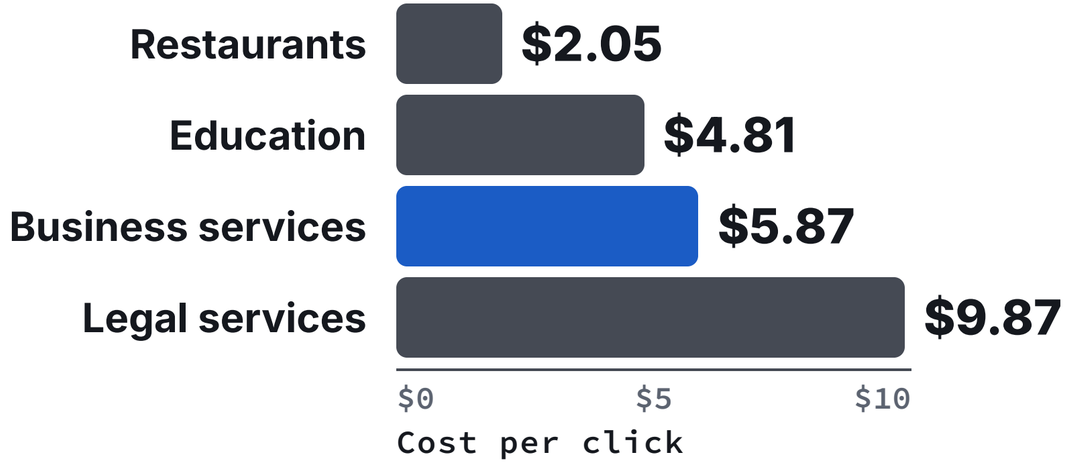
COLD TRAFFIC · WHAT A CLICK COSTS

Clicks cost cents to dollars. €100 buys a fast, honest test

BY CHANNEL



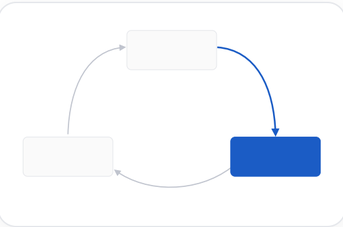
GOOGLE SEARCH, BY INDUSTRY



€100 buys over 100 clicks on Meta, about 20 on Google search, in days. Strangers owe you nothing, so they are honest.

Own row, never toward your bar

DATA LocaliQ, Search Advertising Benchmarks, updated 1 June 2026. Gupta Media, Social CPM Tracker, December 2025, per link click. Dreamdata, LinkedIn Ads Benchmarks 2026, B2B, € drawn as $. Different methods, read them as a sense of scale. Logos are trademarks of their owners.



TAGGED LINKS

Tag every link, or you can't tell who converts

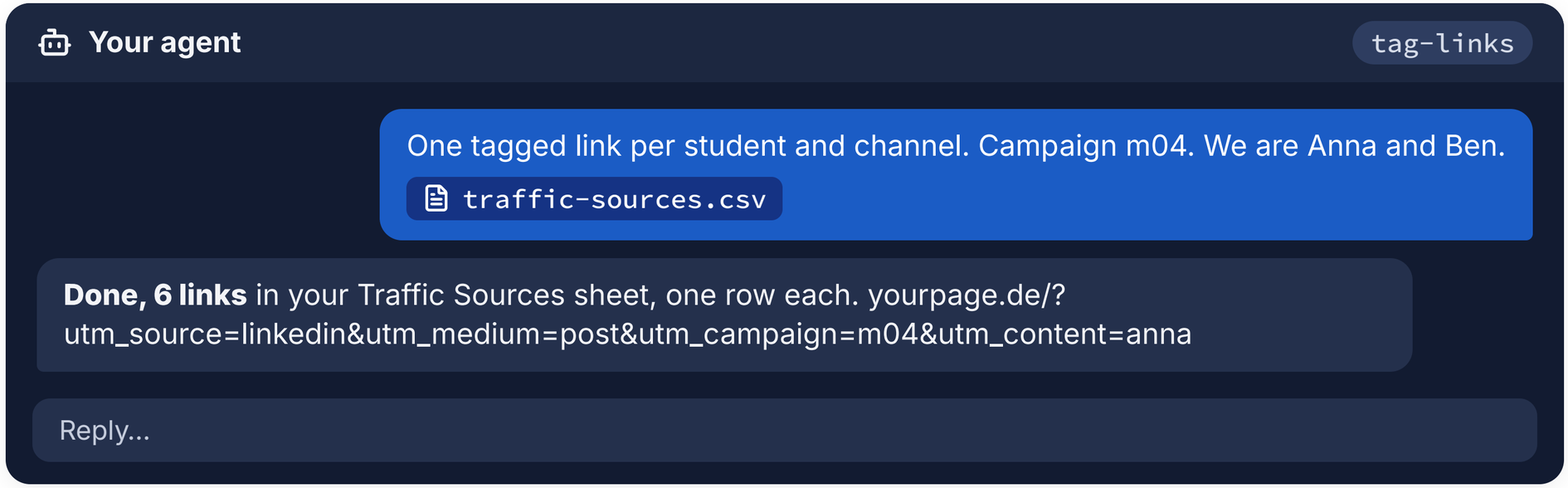
```
yourpage.de/?utm_source=linkedin&utm_medium=post&utm_campaign=m04&utm_content=anna
```

Where | **How** | **Which test** | **Who sent it**

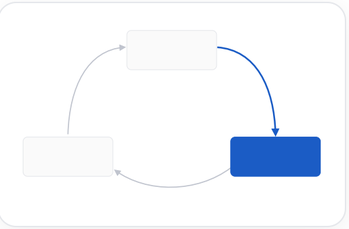
DATA Google, Analytics Help, URL builders: collect campaign data with custom URLs, as of October 2026. Example link illustrative.

WITH YOUR AGENT

Tag every link, or you can't tell who converts



DATA Google, Analytics Help, URL builders: collect campaign data with custom URLs, as of October 2026. Example link illustrative.



HOW-TO · ANALYTICS · **Microsoft Clarity** · STEP 1 OF 4

Set up the dashboard before the first visitor

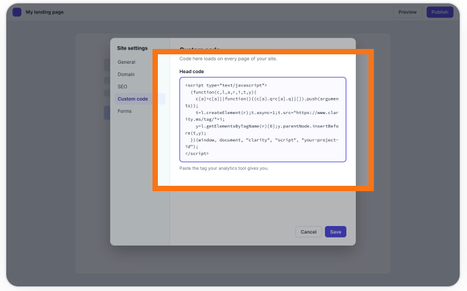
Code here loads on every page of your site.

Head code

1

```
<script type="text/javascript">
  (function(c,l,a,r,i,t,y){
    c[a]=c[a]||function(){(c[a].q=c[a].q||[]).push(argumen
ts)};
    t=l.createElement(r);t.async=1;t.src="https://www.clar
ity.ms/tag/"+i;
    y=l.getElementsByTagName(r)[0];y.parentNode.insertBefo
re(t,y);
  })(window, document, "clarity", "script", "your-project-
id");
</script>
```

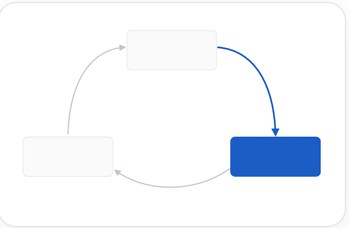
Paste the tag your analytics tool gives you.



1. **Add the tag to your page**
2. Add a consent banner for EU visitors
3. Filter by source
4. Watch your segment's sessions

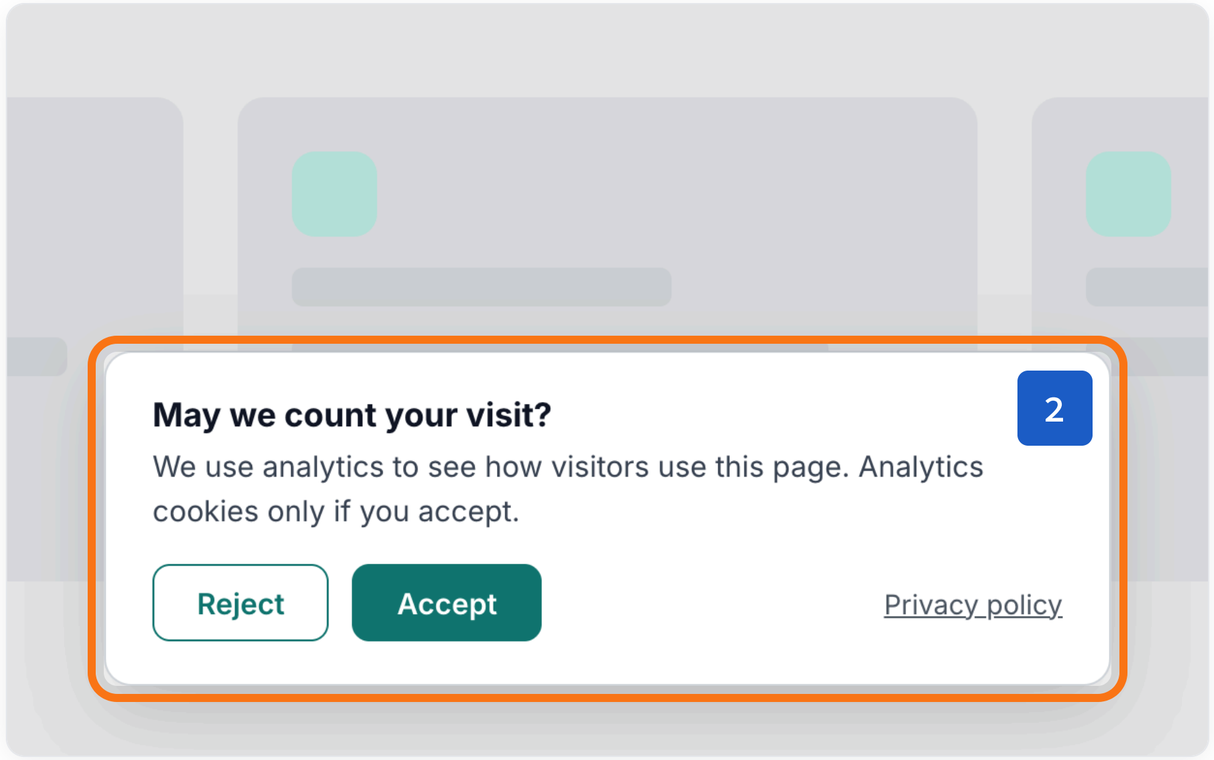
In Clarity, Settings, then Setup, then Install manually

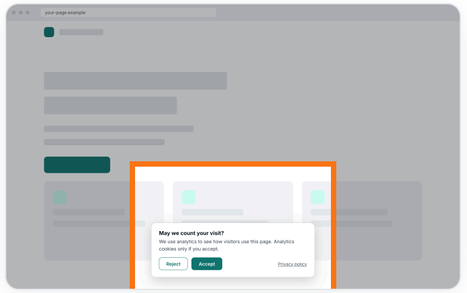
ILLUSTRATIVE Screen prototyped for this deck. Setup path from the Microsoft Clarity documentation, as of October 2026.



HOW-TO · ANALYTICS · STEP 2 OF 4

Add a consent banner for EU visitors

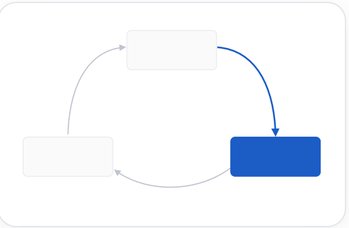




1. Add the tag to your page
2. **Add a consent banner for EU visitors**
3. Filter by source
4. Watch your segment's sessions

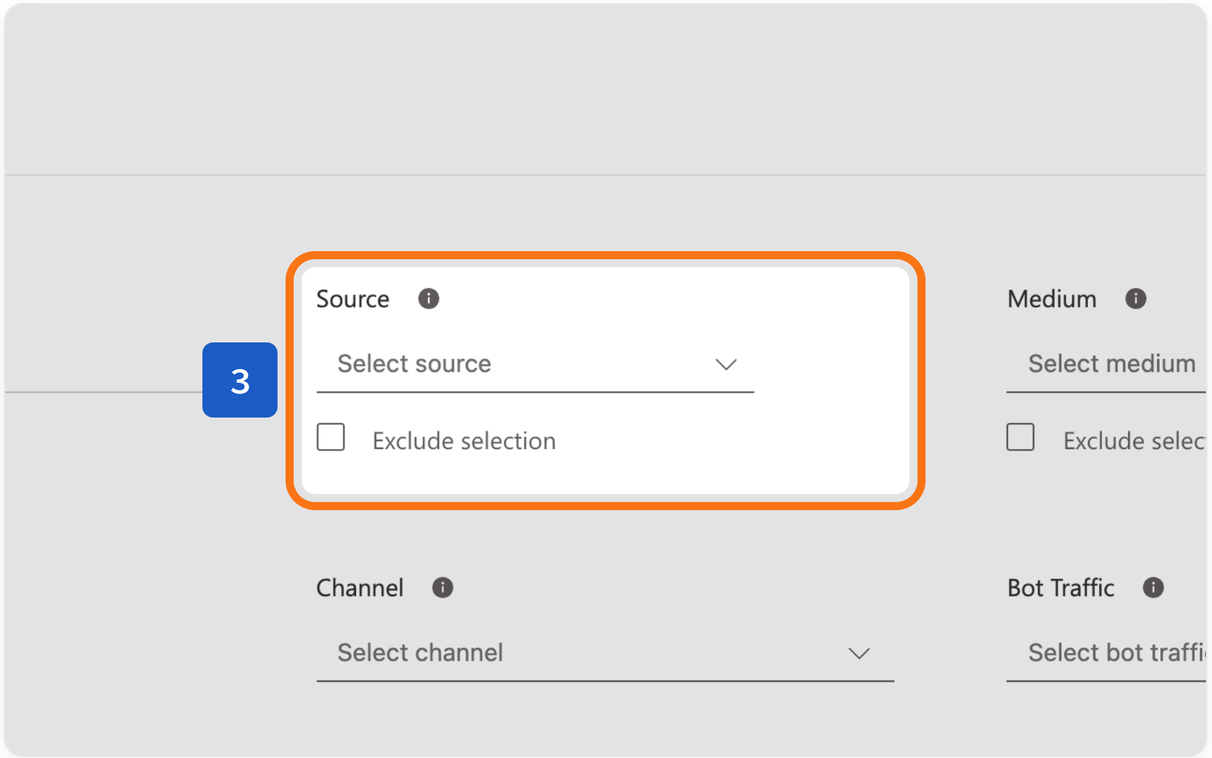
EU visitors: banner first.

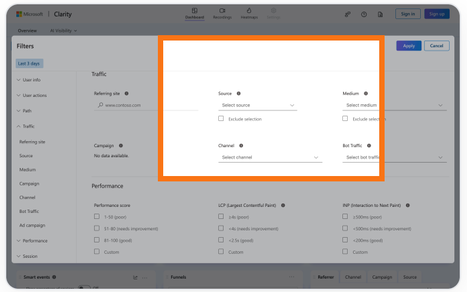
ILLUSTRATIVE Screen prototyped for this deck. Consent rules from the Microsoft Clarity documentation, as of October 2026.



HOW-TO · ANALYTICS · STEP 3 OF 4

Filter by source

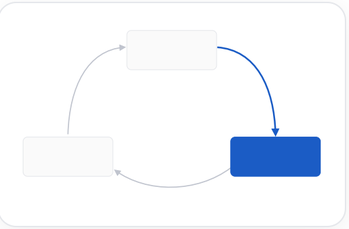




1. Add the tag to your page
2. Add a consent banner for EU visitors
3. **Filter by source**
4. Watch your segment's sessions

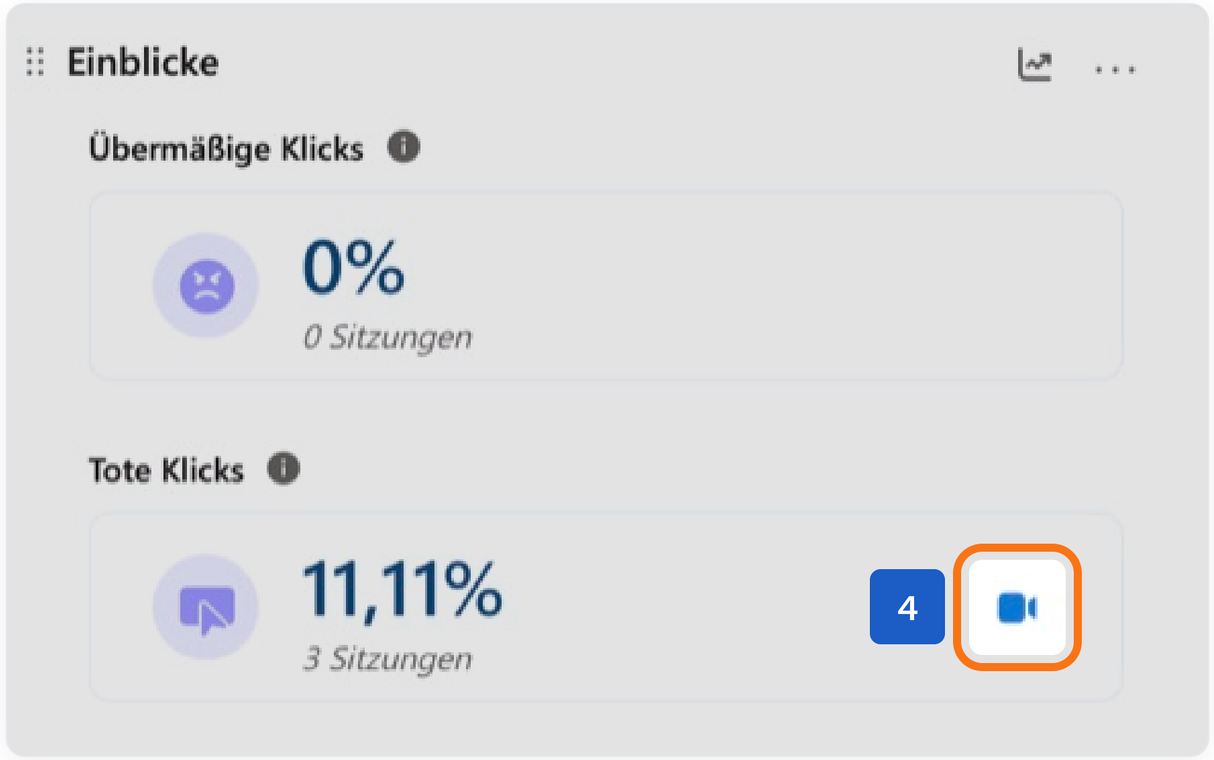
Filters, then Traffic, then Source

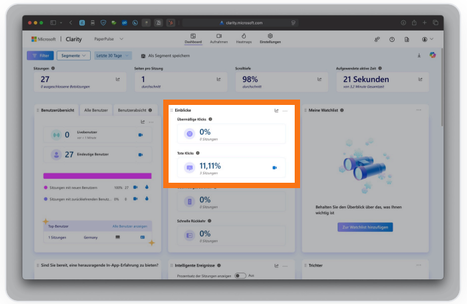
Screenshots of Microsoft Clarity's public live demo as of 26 October 2026, captured with scripts/shots.py.



HOW-TO · ANALYTICS · STEP 4 OF 4

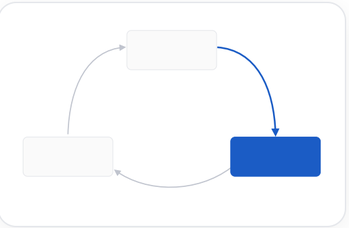
Watch your segment's sessions





1. Add the tag to your page
2. Add a consent banner for EU visitors
3. Filter by source
4. **Watch your segment's sessions**

Tote Klicks are dead clicks, 11% of sessions

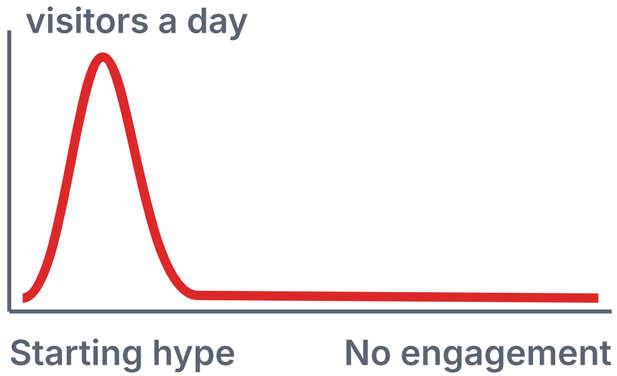


READING YOUR RESULT

Read the curve, then your number

Each curve shows visitors a day, over the weeks after launch

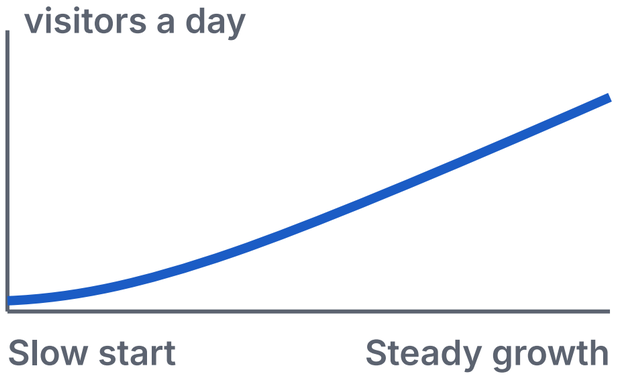
No interest



Wrong group, problem or words

Rework the problem space

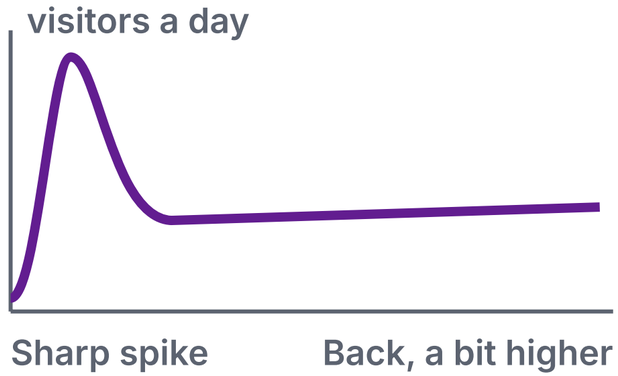
Organic interest



Right group, sustained interest, good words

Find more of them

External shock



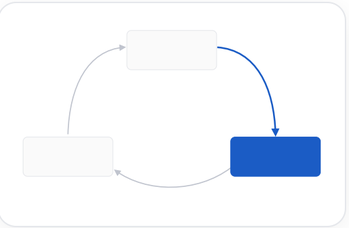
An article or a post went viral

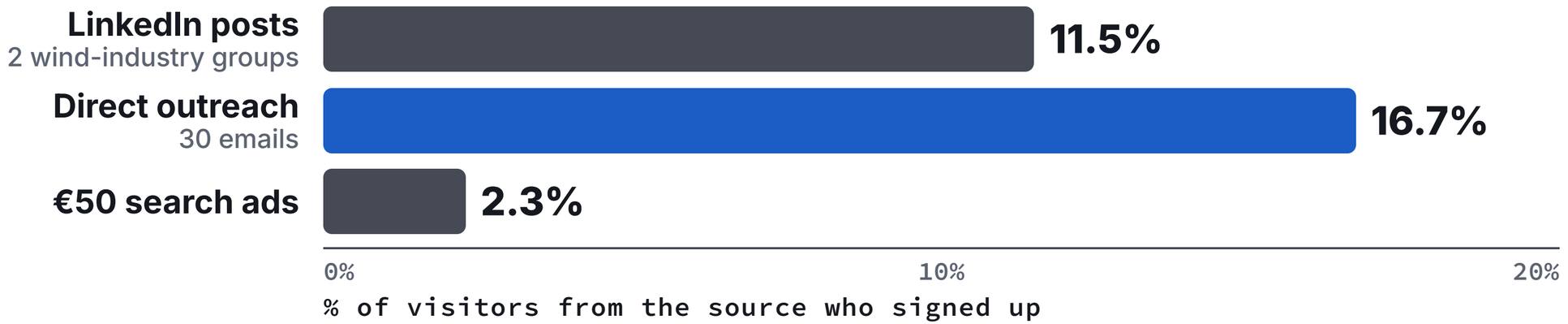
Don't count the spike

Impressions < clicks < sign-ups. Remember the hierarchy of evidence!

DOUBLE DOWN · AN EXAMPLE

Double down where your segment converts





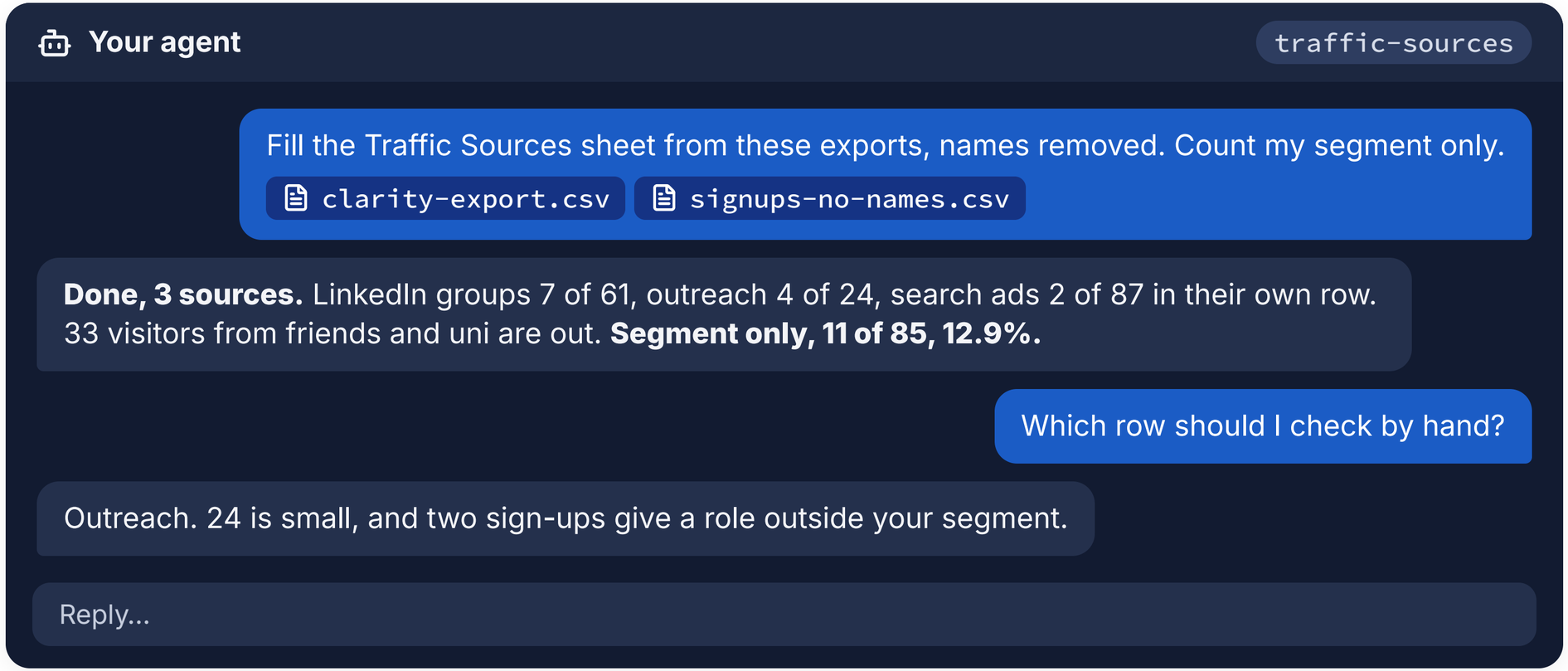
12.9%
segment only, 11 of 85

33
visitors from friends and uni, excluded

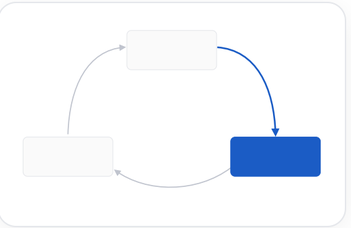
ILLUSTRATIVE Synthetic course exemplar, Landing-Page Test, team TurbineTrack. 4 of 24, 7 of 61, 2 of 87. Bars to scale. The chat is made up too.

WITH YOUR AGENT

Double down where your segment converts



ILLUSTRATIVE Synthetic course exemplar, Landing-Page Test, team TurbineTrack. 4 of 24, 7 of 61, 2 of 87. Bars to scale. The chat is made up too.



NOW YOU · AS A TEAM · 3 MINUTES

Your traffic math

Sign-ups How many do you want?

Share What do you expect?

Visitors So how many?

Places Name three, with people

3:00

OUTPUT

Three named places

INPUT 4 OF 4 · COUNT IT · THE ANSWERS

Count it: your segment, by source

1

How many visitors do you need?

Work backwards from the sign-ups you need: at least 15 visitors a day each

2

Where do you find them?

Where your segment already is. Name the room, not the channel

3

How do you read the result?

Tag every link, count only your segment, read the curve, then your number

CHAPTER 7 OF 9

Wrap-up and check-out

What you take into the week, and how today went for you.

01 ~~Previously on Ambitious Entrepreneurship~~

02 ~~Your presentations~~

03 ~~Rank it: what your evidence proves~~

04 ~~Draw it: the line before the data~~

05 ~~Launch it: a page that tests your problem~~

06 ~~Count it: visitors from your segment~~

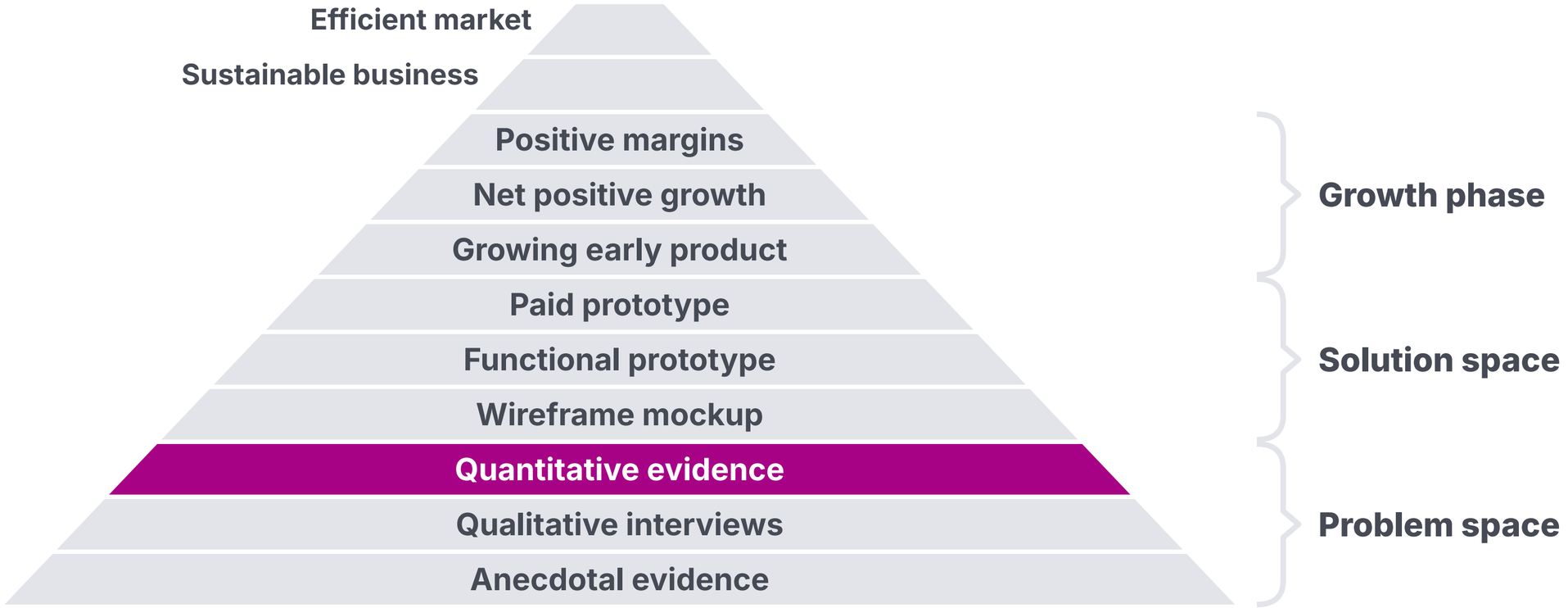
07 **Wrap-up and check-out**

08 Workshop: launch a page that tests your problem

09 Case discussion

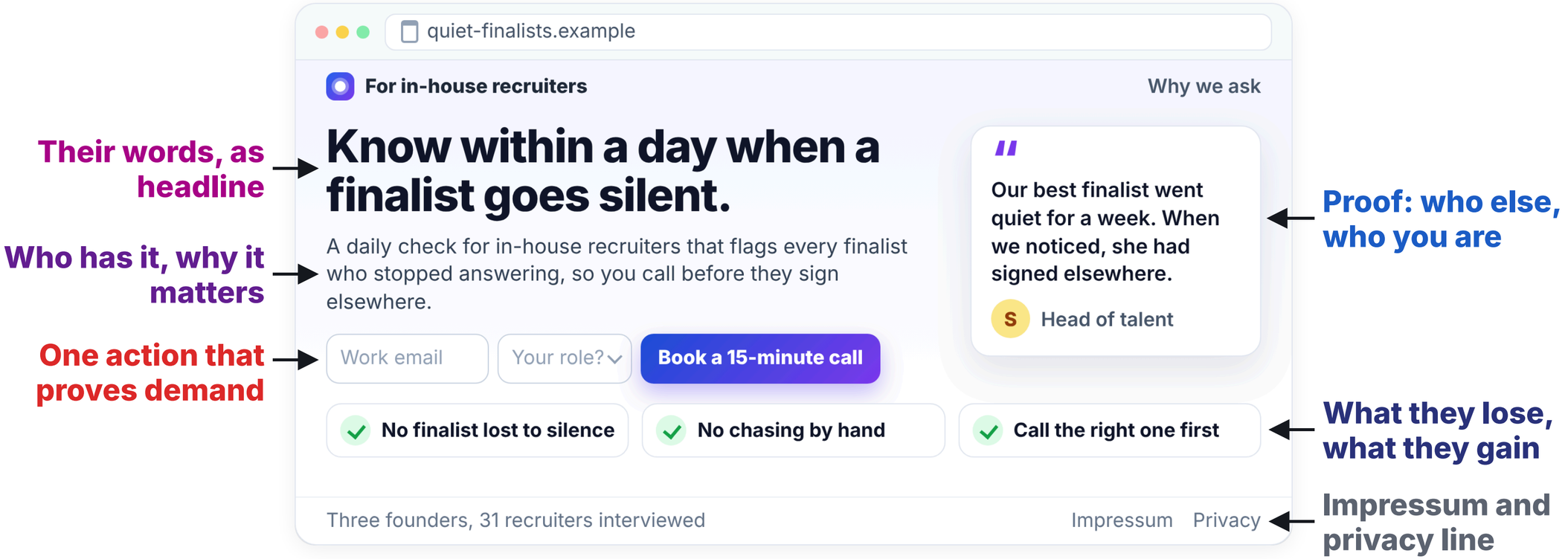
RECAP · STEP THREE IS A SURVEY OR A PAGE, 20+ PEOPLE

From today you climb from saying to doing



RECAP · YOUR PAGE

Their problem, their words, one button



ILLUSTRATIVE Page prototyped for this course, the made-up recruiting team of the workshop.

REFLECTION 1 OF 3 · THE CONTENT

What could you apply?



What could you apply?



What gave you a hard time?

REFLECTION 2 OF 3 · THE PROCESS

Man vs. machine



Where did AI actually help you?



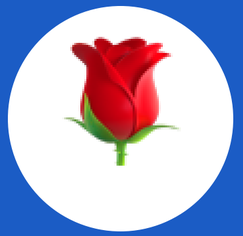
Where was your genius needed?



What are your best hacks to share?

REFLECTION 3 OF 3 · THE RESULTS

Rose, thorn, bud



What was the breakthrough today?



What was the hardest part today?



What is an idea/feature you didn't have time to finish, but want to explore later?

HOMEWORK UNTIL NEXT TIME

Publish one landing page per team, log its pass or fail line before the first visitor arrives, and bring 100 visitors per student from your segment to it.

100 visitors each

The learning floor is the live page, the logged line and 30 visitors per student from your segment.

Your line is logged before anyone visits.

HOMEWORK · HOW TO DO IT

Seven moves for the week

1. Start with your Problem space (M03) prospects
2. Go where your segment is
3. Tag every link
4. Never move the line, log any copy change
5. Call three sign-ups each
6. Survey your sign-ups, optionally
7. Get a blind second opinion

HOMEWORK · WHAT GOOD LOOKS LIKE

A good week ends on your line, not on a feeling

	GOOD	WEAK
Line	10%, logged before launch	Written after the first numbers
Traffic	140 tagged visitors from two recruiter groups	400 visitors, mostly friends
Result	9 of 140, under the line, segment switched	"Lots of interest!"

ILLUSTRATIVE Made-up example from the Validating problems (M04) homework, the recruiting team of the workshop.

Sources in this deck

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See you next time!

Workshop: Launch a page that tests your problem

Write the test, launch the page, and send your first visitors before you leave.

What evidence would change your mind?

01 Previously on Ambitious Entrepreneurship

02 Your presentations

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 **Workshop: launch a page that tests your problem**

09 Case discussion

WORKSHOP · STEP 1 OF 3

Write the test, then the copy

1 STEP 1 · 30 MIN

WRITE THE TEST, THEN THE COPY

Decide what result would change your mind

- → We believe [segment] will [action] when we promise [outcome]
- → Kill line in the Kill Criteria Log, dated, before the page exists
- → Raw notes into an AI, names removed, five hero sections
- → Pick the headline closest to what someone said

Output Test card, dated kill line, hero section

Minutes 30



30:00

WORKSHOP · STEP 2 OF 3

Launch in under an hour

2 STEP 2 · 35 MIN

LAUNCH IN UNDER AN HOUR

A live page that names their problem, asks for one action

- → Their words, why it matters, what changes, one button
- → Form: email plus one segment question
- → Impressum and privacy line, or it stays offline
- → Analytics with visitors and sign-ups per source, then publish
- → Five-second test with another team

Output A public link and analytics that count

Minutes 35



35:00

WORKSHOP · STEP 3 OF 3

Bring your segment, then commit

3 STEP 3 · 25 MIN

BRING YOUR SEGMENT, THEN COMMIT

Leave with visitors on their way

- → Work backwards: 10 sign-ups at 10% need 100 visitors
- → Prospects first, then communities, groups, newsletters
- → One tagged link per student and channel
- → First messages sent before you leave
- → Each student picks floor or full bar, midweek pace

Output Tagged links, sent messages, targets, pace

Minutes 25

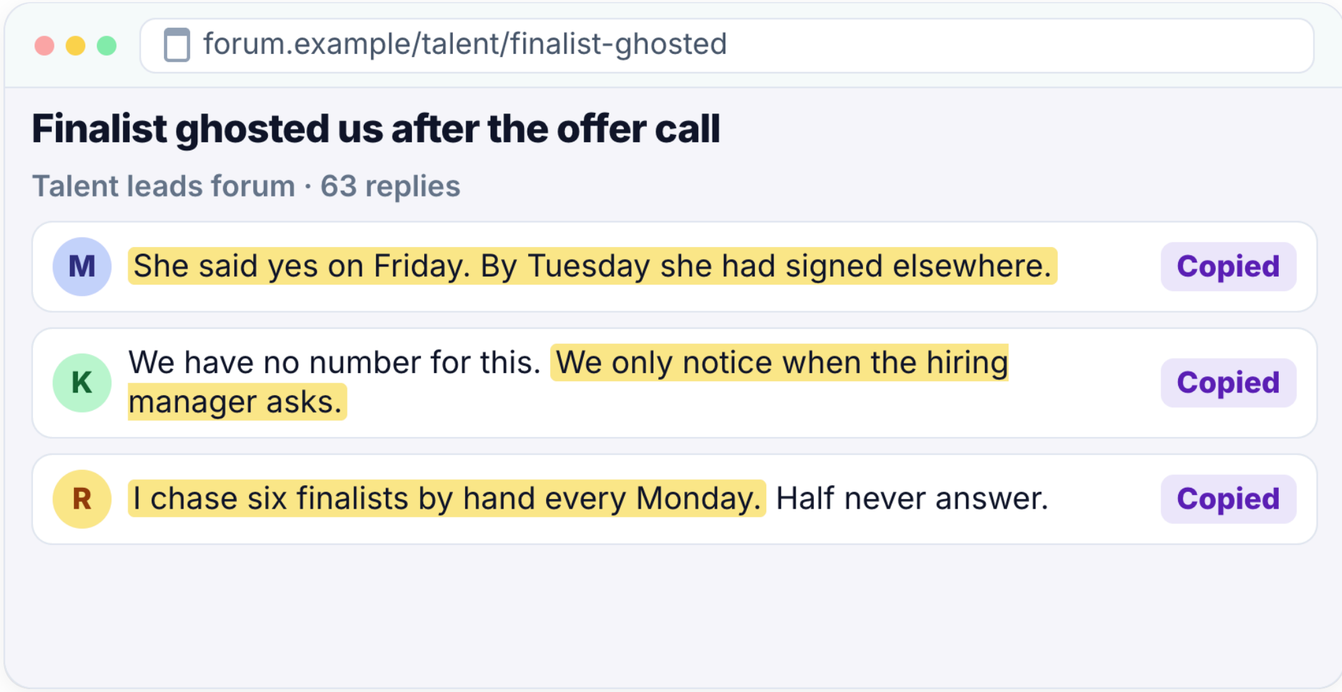


25:00

ARRIVED EMPTY-HANDED?

Start in the forums where they complain

No interview notes? Copy your segment's exact sentences from the forums where it complains, mark the promise as assumed, and post your link there in Step 3.



ILLUSTRATIVE Screen prototyped for this course, forum and replies made up. Validating problems (M04) workshop.

HOMEWORK UNTIL NEXT TIME

Publish one landing page per team, log its pass or fail line before the first visitor arrives, and bring 100 visitors per student from your segment to it.

100 visitors each

The learning floor is the live page, the logged line and 30 visitors per student from your segment.

Your line is logged before anyone visits.

See you next time!

MODULE 04 · VALIDATING PROBLEMS · CASE DISCUSSION · ABOUT 90 MINUTES

Case discussion: Fairphone

Stand in Fairphone's shoes on 13 May 2013 and decide whether to sell 5,000 phones before making one.

Do you open the pre-sale on this benchmark?

01 Previously on Ambitious Entrepreneurship

02 Your presentations

03 **Rank it: what your evidence proves**

04 **Draw it: the line before the data**

05 **Launch it: a page that tests your problem**

06 **Count it: visitors from your segment**

07 Wrap-up and check-out

08 Workshop: launch a page that tests your problem

09 **Case discussion**

CASE · Fairphone · AMSTERDAM, 2010 TO 13 MAY 2013

Three years of campaign, one month left to sell 5,000



DATA Validating problems (M04) case, timeline and Exhibit 1. Nothing after 13 May 2013.

CASE · MONDAY 13 MAY 2013 · VOTE 1 OF 2

Open the pre-sale on this benchmark, or not yet?

OPTION A

Open the pre-sale on this benchmark

OPTION B

Not yet

€325
the price, paid up front

5,000
phones sold

15 June
2013, the deadline

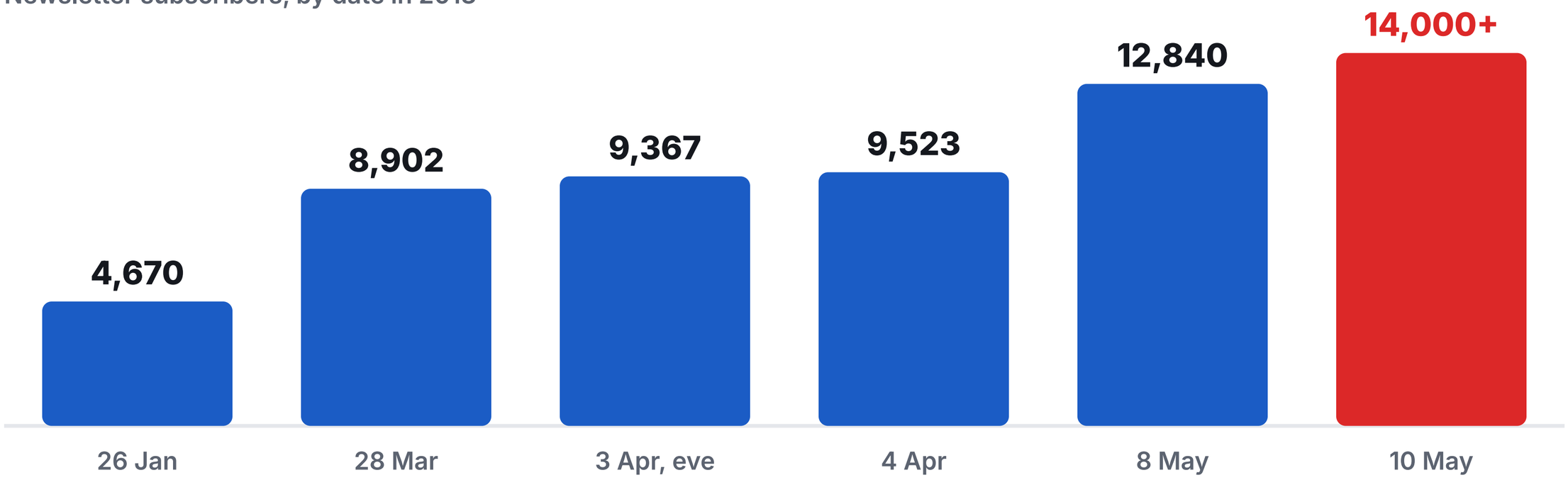
Refund
if the orders are cancelled

DATA Validating problems (M04) case, Exhibit 2 (shop terms, articles 4.3 and 4.4, adopted 13 May 2013) and Exhibit 3.

CASE · EXHIBIT 1 · NEWSLETTER SUBSCRIBERS, 2013

What do 14,000 sign-ups prove?

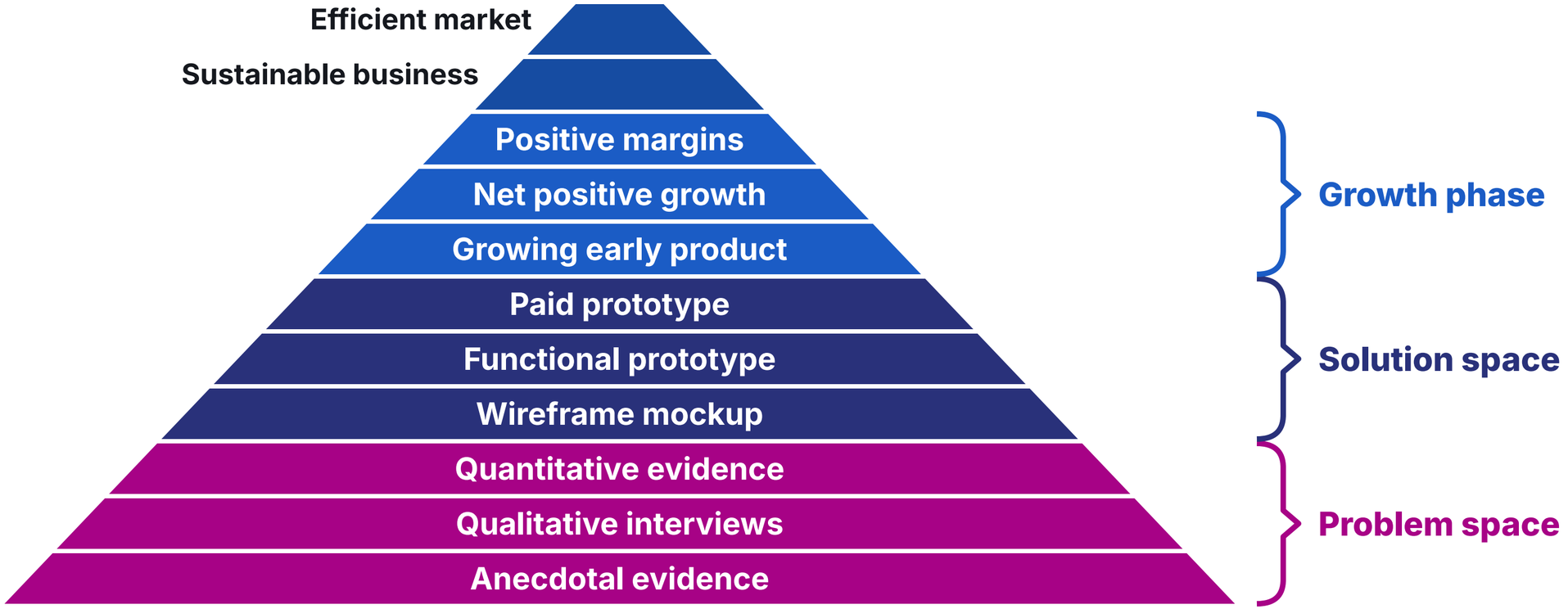
Newsletter subscribers, by date in 2013



DATA Validating problems (M04) case, Exhibit 1: archived newsletter counter, Fairphone blog of 4 April 2013 (company figures for the evening of 3 April and for 4 April), Telecompaper of 10 May 2013 ("more than 14,000", drawn at 14,000). Bars to scale, 0 to 15,000.

CASE · DIAGNOSIS

Which step does each piece stand on?



Fairphone has a sales benchmark of 5,000 (five thousand) mobile phones, which must be sold before the 15th of June 2013.

Shop terms, article 4.3, adopted 13 May 2013. If Fairphone cancels on a miss, the buyer gets "a full refund of the money the Customer already has paid" (4.4)

DATA Validating problems (M04) case, Exhibit 2: Fairphone B.V., Customer Terms and Conditions, articles 4.3 and 4.4, adopted 13 May 2013.

CASE · BOARD PLAN

Is 5,000 by 15 June a good test of demand?

	Vote 1	Vote 2	Prediction
Evidence on 13 May and its step	What, and which step?		
What it proves and what it doesn't	Proves · doesn't prove		
The line as a test	Count · threshold · date Pass · miss		
5,000 buyers from where?	Share of the list, then?		

CASE · VOTE 2 OF 2

Same question. Who moved, and why?

OPTION A

Open the pre-sale on this benchmark

OPTION B

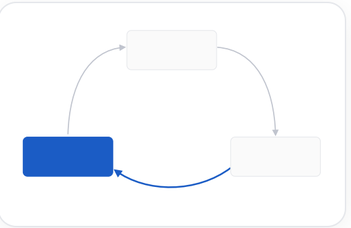
Not yet

€325
the price, paid up front

5,000
phones sold

15 June
2013, the deadline

Refund
if the orders are cancelled



CASE · BRIDGE

Before your page exists, write your line

Fairphone wrote its line into its terms before the first buyer arrived.

We are right if	[X]% or more by [date], logged today
Then	pass: [next step] · fail: switch segment or problem



FRAMEWORK The last two rows of the test card, after Bland and Osterwalder (2019), as on slide 30. Photo: StartupStockPhotos on Pixabay.

See you next time!